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An aerodynamic approach to the migration of the Short-toed Snake Eagle (*Circaetus gallicus*) over the Sahara Desert

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Research article

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Résumé

Mots-clés : aéroécologie, migration, Sahara, *Circaetus gallicus*, télémétrie GPS haute fréquence

La migration longue distance des oiseaux relie leurs sites de reproduction et d'hivernage via deux modes de vol, le vol battu et le vol plané. La traversée de zones critiques est exigeante chez les planeurs, nécessitant des aptitudes généralement apprises en suivant des conspécifiques. L'ontogenèse des migrateurs planeurs solitaires est ainsi mal comprise. Nous étudions l'effet de l'expérience et de la saison sur les performances de vol, leur optimisation, et les comportements liés au vent chez le Circaète Jean-le-Blanc (*Circaetus gallicus*) lors de la traversée du Sahara, à partir de données de télémétrie GPS haute fréquence, couvrant 46 migrations (2020–2026) de 13 individus (4 expérimentés, ≥ 5 ans ; 9 inexpérimentés, ≤ 3 ans). L'efficacité de la route migratoire n'a pas été influencée par l'expérience ; seule la saison montrait une tendance sur la durée de traversée (6,1 ± 3,2 vs 6,8 ± 2,2j en automne, 7,5 ± 3,5 vs 11,2 ± 2,5j au printemps). Les métriques de vol ont montré une interaction marquée expérience×saison : la hauteur de vol atteignait 713 ± 115 m chez les expérimentés contre 565 ± 80 m chez les inexpérimentés en automne, mais 490 ± 100 m contre 610 ± 65 m au printemps. Un schéma similaire a été observé pour le gain d'altitude, la vitesse et la durée des thermiques, s'expliquant par la migrations précoce des expérimentés au printemps pour la reproduction. La fréquence de prise de thermiques et l'optimisation du vol par ajustement de la vitesse à la force estimée du prochain thermique, révélaient une utilisation modérée mais précocement maîtrisée des ascendances. La compensation restait le comportement dominant face au vent, plus marquée près des côtes, sans effet de l'expérience. La saison, plutôt que l'expérience, s'impose en principal facteur déterminant, les inexpérimentés égalant les expérimentés, suggérant des compétences innées plutôt qu'acquises.

Abstract

Keywords : aeroecology, migration, Sahara, *Circaetus gallicus*, high-frequency GPS telemetry

Long-distance bird migration connects breeding and wintering sites via two flight modes, flapping flight and soaring-gliding flight. Crossing critical areas is demanding for soaring migrants, requiring skills learned by following conspecifics. The ontogeny of solitary soaring migrants is thus poorly understood. Here we investigate the effect of experience and season on flight performance, its optimization, and wind-related behaviors in the Short-toed Snake Eagle (*Circaetus gallicus*) during Sahara crossing, using high-frequency GPS telemetry data covering 46 migrations (2020–2026) from 13 individuals (4 experienced, ≥ 5 years; 9 inexperienced, ≤ 3 years). Migratory route efficiency was not influenced by experience; only season showed a NS trend on duration and distance (6.1 ± 3.2 vs 6.8 ± 2.2 days in autumn, versus 7.5 ± 3.5 vs 11.2 ± 2.5 days in spring). Flight metrics showed a marked experience×season interaction: flight height reached 713 ± 115 m in experienced birds versus 565 ± 80 m in inexperienced birds in autumn, but 490 ± 100 m versus 610 ± 65 m in spring. A similar pattern was observed for altitude gain, speed and thermal duration, explained by the breeding pressure experienced birds face in spring. Thermal uptake frequency and flight optimization through speed adjustment to the estimated strength of the next thermal revealed moderate but early-mastered use of updrafts. Compensation remained the dominant behavior in response to wind, more pronounced near coastlines, with no effect of experience. Ultimately, season rather than experience was the main source of apparent performance differences, with inexperienced birds matching experienced ones, suggesting innate rather than acquired skills.

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I. Introduction

Long-distance migration is one of the main characteristics of birds and has been the subject of intensive research on its ecology and evolution (Newton 2023). Migration refers to the seasonal movements carried out each year by many animal species, particularly birds, between their breeding and wintering areas (Newton 2023). This mechanism allows for the optimization of resource use according to seasonal variability, for the reduction of competition and predation pressure, and contributes to maintain physical condition (Cox 1968; Eisaguirre *et al.* 2019; McKinnon *et al.* 2010; Nathan *et al.* 2008; Somveille *et al.* 2015). This type of movement, driven by phenological responses triggered by photoperiod (Bani Assadi & Fraser 2021), leads migratory birds to follow well-defined routes, or migratory flyways, that may extend from hundreds to several thousand kilometers across the continents (La Sorte & Fink 2017; Newton 2023). Migration represents a very critical phase in the life cycle of raptors, characterized by high mortality, especially among juveniles (Strandberg *et al.* 2010). More than half of the annual deaths of migratory birds occurs during migration, with mortality often higher in spring than in autumn, particularly for populations crossing the Sahara desert (Lok *et al.* 2015). The Sahara is the area with the highest mortality (Klaassen *et al.* 2014; Strandberg *et al.* 2010), due to its complex topography, including dunes and ergs, and strong winds blowing from the northeast, known as the Harmattan, which constitute a major barrier for migratory birds (Biebach 1990; Vansteelant *et al.* 2017b). Crossing areas devoid of resources (oceans, deserts) indeed represents a demanding challenge to migratory birds, involving trade-offs between crossing speed and the exploitation of encountered conditions (Alerstam 2001). To deal with those difficulties, long-distance migratory birds can operate two different types of flight types: flapping, which is highly energy-demanding (Zenzal *et al.* 2018), and gliding, less energy-demanding but dependent on weather conditions and thermal updrafts (Pennycuick 1972). Depending on their reliance on these two modes, migratory birds can be classified into three categories: strict flappers, mostly small passerines unable to exploit thermal updrafts and thus reliant almost exclusively on active flapping flight; mixed flyers, which alternate between flapping and gliding, and strict gliders (Brown 1963). Strictly soaring birds, such as the Griffon Vulture, take advantage of thermal updrafts, enabling them to save more than 60% of their energy compared to flapping flight during long-distance travel (Duriez *et al.* 2014). During soaring, birds gain altitude when circling in these thermal convective updrafts (commonly referred to as thermals: localized regions of rising, buoyant air heated by sunlight (Predelli 2023)), which subsequently enables them to glide for horizontal displacement (Cone 1962;

Spaar & Bruderer 1996; Williams 1990). Soaring flight is a learned and acquired behavior (Abrahms *et al.* 2019; Harel *et al.* 2016), requiring advanced cognitive skills to locate uplifts as well as fine-tuned locomotor skills for optimal adjustment of the body and wings to extract the most energy from them (Williams *et al.* 2018). This alternation between thermal climbs and glides between successive thermals is known as cross-country soaring flight, and allows birds to cover long distances (Duriez *et al.* 2014; Pennycuick 1972). During the glide phase, birds face a trade-off between glide angle and speed: a shallow angle results in the minimum sink rate, the horizontal speed at which the least altitude is lost per unit time, though not necessarily the best-glide condition, which instead maximises the distance covered per unit of altitude lost, whereas a steeper angle allows for a higher speed at the cost of a faster loss of altitude, a strategy that only pays off if the next thermal is strong enough to quickly regain the altitude lost (Horvitz *et al.* 2014; Rayner 1990). Optimising the use of thermals and the subsequent glides, notably by adjusting gliding speed in anticipation of the strength of the next thermal, therefore makes it possible to maximise overall travel speed. MacCready's theory formalises this optimal balance by predicting the gliding speed between thermals, based on the strength of the next expected thermal (Cochrane 1999; Keskin *et al.* 2025). Beyond this fine-scale optimization, wind more broadly is a major factor to master during migration, acting as either an ally or an obstacle, as wind speeds are of the same order of magnitude as birds' airspeeds, flight costs can easily be doubled or halved by wind effects (Liechti 2006). Migratory birds must therefore also adapt their behavior and route accordingly, balancing the minimisation of energy expenditure against the minimisation of time and distance spent in critical zones (Liechti 2006; Panuccio *et al.* 2012; Richardson 1990). Because wind can push them off their intended heading, individuals may either drift with it, accepting lateral displacement, or compensate by adjusting their trajectory to maintain a more direct course (Vansteelant *et al.* 2017b, a).

This combination of skills can be acquired through social learning, based on parental guidance (Byholm *et al.* 2022; Kölzsch *et al.* 2020; Méndez *et al.* 2021), on conspecifics, in species that migrate in groups such as White Storks (*Ciconia Ciconia*) (Brønnvik *et al.* 2024; Rotics *et al.* 2016), or independently, through individual experience, in solitary migrants (Newton 2024). In the Short-toed Snake Eagle (*Circaetus gallicus*) (Gmelin 1788), a strict trans-Saharan migrant, juveniles migrate independently, without guidance, and complete a full migration during their very first journey (Agostini *et al.* 2017; MALAFOSSE & MAIGRE 2014). In long-lived solitary migratory species such as the Short-Toed Snake Eagle, experience gained over successive migrations can allow for a progressive optimization (Vidal-Mateo *et al.* 2016). Experience has been shown to be a crucial factor in other solitary migrants, adult Marsh Harriers

(*Circus aeruginosus*), for example, react differently to winds from opposing directions, possibly by gauging local conditions relative to what they have learned to expect from previous journeys (Klaassen *et al.* 2011). Flight metrics during migration such as speed, climbing altitude, crossing duration and the efficiency of thermal use, could be directly linked to flight experience, leading to differences between experienced and inexperienced. Such differences have been observed in other soaring species, with adults Griffon Vultures (*Gyps fulvus*) outperforming juveniles under challenging thermal soaring conditions (Harel *et al.* 2016). The ontogeny of migration still remains poorly understood in species that migrate solitarily from their very first migration, and especially for soaring species (Flack *et al.* 2022). The Short-toed Snake Eagle relies almost entirely on soaring flight to optimise energy use, making it a particularly compelling species for studying such optimization strategies, as experience would be expected to play a major role in refining thermal-use efficiency. Also, spring migration, from March to late May (Zenzal & Moore 2016), is subject to strong time pressure, as adults must return quickly to their breeding sites to maximize reproductive success, a constraint that applies less to inexperienced, which do not yet breed (Chan *et al.* 2024; Jiguet *et al.* 2012). In contrast, autumn migration, between September and mid-November, is less time-constrained and corresponds to movement toward wintering grounds in Africa, where conditions are more favorable (Somveille *et al.* 2018).

Onboard GPS technology is currently revolutionizing ornithology, particularly the study of migration. It enables the collection of high-frequency telemetry data on bird movements, providing detailed information on routes, stopover duration, resting behavior, altitude, and flight speed (Soriano-Redondo *et al.* 2020; Treep *et al.* 2016)**.** Indeed, the development of on-board electronics for birds, generating GPS telemetry data, now makes it possible to characterise the flight characteristics of birds very precisely (Keskin *et al.* 2025), particularly during migration (Brønnvik *et al.* 2024). This technological advance offers new opportunities to study flight strategies and their associated costs, which are still largely unknown. Other studies on the subject use 'high-frequency' GPS data, but not at extremely high resolutions such as 1 point every 1-3 seconds, which would allow precise characterization of optimization strategies (Schaub *et al.* 2023). Taking advantage of this level of resolution, we here apply it to the Short-toed Snake Eagle, with our first objective being to characterize flight performances during the Sahara crossings. Specifically, we assess flight performances through the efficiency of migration, in terms of crossing duration, distance, general flight performance metrics (height, speed, altitude gain), and the use of thermal updrafts. We specifically expect that flight metrics will be higher in experienced adults compared to inexperienced individuals. We will also test

the hypothesis that spring migration is more demanding than autumn migration, due to less favorable weather conditions, particularly for adults whose phenology is constrained by reproduction. Adaptations and optimizations of flight strategies at a fine scale, following MacCready's theory, as in Keskin *et al.* (2025) will be tested. We hypothesise that, due to their experience, experienced individuals are better able to adjust their flight speed to the strength of the next thermal, that is, to more accurately anticipate the strength of the upcoming thermal. Finally, regarding wind-adaptive behavior, we hypothesise that experienced individuals show greater drift and less compensation, tolerating lateral displacement because they can better judge when it can be corrected later without added risk, saving energy rather than compensating systematically. We also expect stronger compensation in spring, when conditions are harsher. We further hypothesise that the latitudinal distribution of compensation behavior differs between seasons, with compensation concentrated at the latitudes associated with the riskiest wind regimes, the southern Sahara in spring and the central Sahara in autumn, where the cost of uncorrected drift (e.g. of being pushed out over the Atlantic) is highest.

II. Materials and Methods

II.1 The Short-toed Snake Eagle: Ecology and Migratory Strategy

The Short-toed Snake Eagle is a strictly trans-Saharan migratory raptor (Vidal-Mateo *et al.* 2016), belonging to a genus that is phylogenetically closer to vultures than to eagles (Lerner & Mindell 2005). The genus *Circaetus* originates from Africa. The Short-toed Snake Eagle is the only species of its genus that breeds in the Palearctic region, including Europe, northern Asia, and North Africa (Cauli *et al.* 2022; Pavón *et al.* 2010), and winters in the Sahel (Christie & Ferguson-Lees 2010). It is a long-lived species with delayed sexual maturity: juveniles generally reach maturity only after five years and do not occupy a breeding territory during at least their first three years (Mellone *et al.* 2011). Adults return to their breeding sites between mid-March and mid-April, while departure to wintering grounds occurs in September and October (Bakaloudis *et al.* 2005). Morphologically, the species is among the heaviest long-distance migratory birds (Agostini *et al.* 2017), with a wingspan reaching up to 1.85 meters and a body mass ranging from 1.3 to 2.3 kilograms depending on sex. It avoids long sea crossings and generally migrates alone, although temporary groups of two to fifteen individuals may be observed following local stopovers related to unfavorable weather conditions (Mellone *et al.* 2016). Finally, juveniles remain at the nesting site well after the parents have departed on

migration (Pavón *et al.* 2010). The delay between fledging and departure is approximately forty days (Agostini *et al.* 2017).

II.2 GPS Telemetry Data

Tagged Individuals

The monitored birds were equipped, between 2020 and 2025, with ©Ornitela OT-25 GPS tags, mounted with backpack type of harness. Tags benefited from an elevated solar panel to improve battery charging (mostly taking place when birds are flying) and extending tag lifespan (expected to ca. 4 years). Individuals were classified as inexperienced (≤ 3 years) or experienced (≥ 5 years, breeding adults). Inexperienced individuals were tagged at the nest 45–55 days after hatching (N = 9). Experienced individuals were equipped during their stay in wildlife rehabilitation centers (N = 3), or through mist-net captures (N = 1). Data were stored onboard and offloaded via 3G/4G GSM connection when coverage was available, with recording frequency automatically adjusted according to location and battery consumption, as configured via the Ornitrack control panel.

Basic GPS data processing

In this study, dorsal GPS tags were attached to the birds using a harness. In accordance with the general rule in telemetry, the transmitter weight (25 grams + 2 grams of Teflon harness) did not exceed 3% of the bird's body mass (Vandenabeele *et al.* 2012, 2014). High-frequency data, recorded at very short time intervals (approximately three seconds), provided that battery level exceeded 50%, were collected within designated areas called "Geofences" (Fig.1). These zones, extending over approximately 100 km from north to south, allow precise measurement of speed, altitude gain, and height above ground level, since the continuous high-frequency recording keeps the GPS module permanently powered on, substantially improving positional accuracy in both the horizontal (x-y) and, especially, the vertical (z) dimension (Bouten *et al.* 2013; Schaub *et al.* 2023). Outside these zones, the recording frequency was reduced to one location every five minutes (or longer intervals when battery levels were low) in order to preserve tag autonomy.

II.3 The saharan zone

The Saharan zone of interest was delineated using the "Ecoregions 2017 ©" (2026) shapefile (Dinerstein *et al.* 2017), retaining only the desert biome crossed during migration (aridity index < 0.05; (Ward 2016)) (Fig. 1). Its southern boundary corresponds to the last geofence band, while its northern boundary follows the desert biome limit as defined by this aridity threshold.

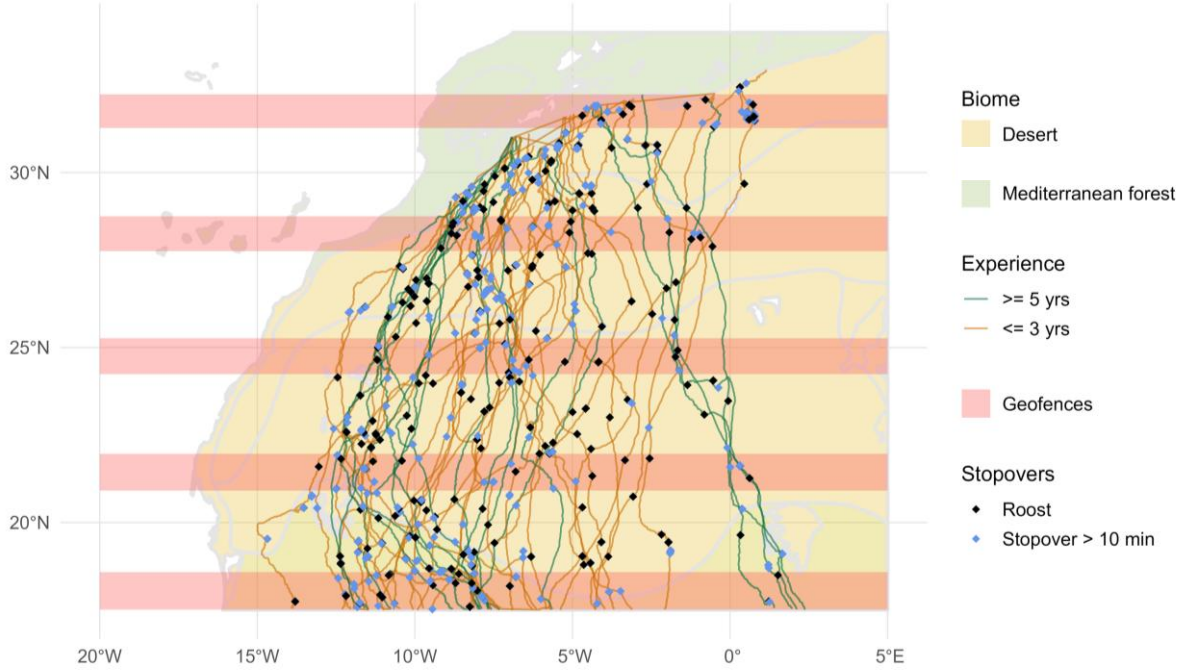


Figure 1. Map of the study area (20°W - 5°E ; 17,5°N - 34°N) showing all-season trajectories of experienced (≥5 years, green) and inexperienced (≤3 years, orange) individuals, with Sahara-crossing geofences shown in red, overlaid on biomes from the "Ecoregions 2017 ©" (2026) dataset. Black diamonds indicate roosting sites and blue diamonds indicate stopovers longer than 10 minutes (no movement, low altitude). North oriented upward; scale 1:15,000,000.

II.3 Data analysis

Basic data processing

Raw GPS data were all downloaded from the Ornitela portal ("OrniTrack Control Panel" 2026) and records were restricted to periods of actual tag deployment. Positions were classified into two acquisition-frequency categories: High Frequency (HF, interval ≤ 6 s between consecutive positions) and Low Frequency (LF, interval ≥ 300 s). Day/night status was determined from solar angle, using two thresholds (−6° and 0°). Altitude above mean sea level was calculated by applying the EGM96 geoid to the ellipsoidal height provided by the tags, while height above ground level was obtained from the SRTM digital elevation model (3 arc-second resolution, approximately 90 m at the equator). In-flight positions were distinguished from ground positions using GPS ground speed thresholds (1.75 m/s for HF data, 2.75 m/s for LF data). A second version of this classification incorporated the additional criterion "HF position more than 50 m above ground = in flight," and was used to retain only in-flight positions for the analysis of flight performances. Subsampling was then applied to maintain a minimum interval of 5 minutes between successive positions (30 s tolerance), yielding three datasets: the full dataset, the dataset subsampled at 5 min, and a dataset restricted to HF data only. Altitude gain

relative to the previous GPS position was also calculated as the positive height difference, with negative values set to zero. For HF data, contiguous segments (with no gap > 6 s) were identified and numbered into individual tracks and those with fewer than 100 positions were excluded. Track durations were highly variable with a majority of shorts tracks [Appendix 1].

Tortuosity, duration and distance

Migration routes were traced from both HF and LF data (using the dataset subsampled at 5 min intervals). Tortuosity was quantified as a straightness index: $T = d_rp/d_vo$ (Rueda-Uribe *et al.* 2022). D_rp is the summed distance between roosts, and d_vo is the shortest (bird-flight) distance between the first and last position within the zone in question. The distance d_vo was determined from GPS data and measured between the entry and exit points of the zone. Thus, when this index equals 1, individuals followed the most direct route. Conversely, the higher the index, the more individuals deviated from the most direct trajectory. Migration duration and distance were calculated over a restricted zone, delimited between the southern edge of the southernmost geofence and the northern edge of the second-to-last geofence, so as to cover only the critical stretch of migration common to all individuals. Duration corresponds to the time interval between the dates on which birds crossed the entry and exit latitudes delimiting this zone (Fig. 1). Distance was calculated as the sum of distances between successive daytime GPS positions, every 5 minutes, within this same zone (also using the dataset subsampled at 5 min).

Flight performances

For all flight performance analyses, only the dataset restricted to high-frequency (HF) data was used. An autocorrelation analysis (based on the autocorrelation function, ACF) revealed pseudoreplication in height above ground level and speed up to 2.87 min and 1.49 min, respectively, corresponding to the time lag beyond which values became statistically independent [Appendix 2]. Subsampling was therefore applied at these time intervals within the high-frequency tracks. Tracks shorter than 5 minutes were excluded, and those longer than 30 minutes were segmented into 30-minute portions. Altitude gain for each track was then extrapolated to 30 min by proportionality [Appendix 3]. For flight behavior identification, GPS positions were differentiated to obtain per-point horizontal and vertical velocities and accelerations, allowing computation of a curvature index $K = (vx \cdot ay - vy \cdot ax) / (vx^2 + vy^2)^{1.5}$. Thermalling climbs were defined as having a curvature $|K| > 0.01$ m^{-1} and a vertical speed $vz >$ 0.3 m/s, following Keskin *et al.* (2025) [Appendix 4]. Two thermal climbs separated by less than 100 m of gliding flight (losing height) were considered as a single thermalling event, and climbs of less than 130 m were excluded. Thresholds were adjusted by comparison with manual

counts, independently assessed from flight trajectories over 6 segments of 1 h and 13 segments of 2 h [Appendix 5]. Finally, the first hour after sunrise was excluded, and a sinuosity index, defined as the ratio of total distance to net displacement per track, was computed to identify and exclude flight-and-forage behavior (sinuosity > 2.3).

Cross Country Optimization strategy

Cross-country flight alternates between thermalling (circling within updrafts to gain altitude) and gliding (straight, powerless descent), as illustrated in the 3D flight trajectory (thermals in orange/red, glides in blue; Fig. 2). Cross-country optimization strategy describes how efficiently individuals exploit this thermal-glide sequence, following MacCready theory, whereby optimal flight speed and glide angle depend on anticipating the strength of the next thermal to adjust glide performance accordingly.

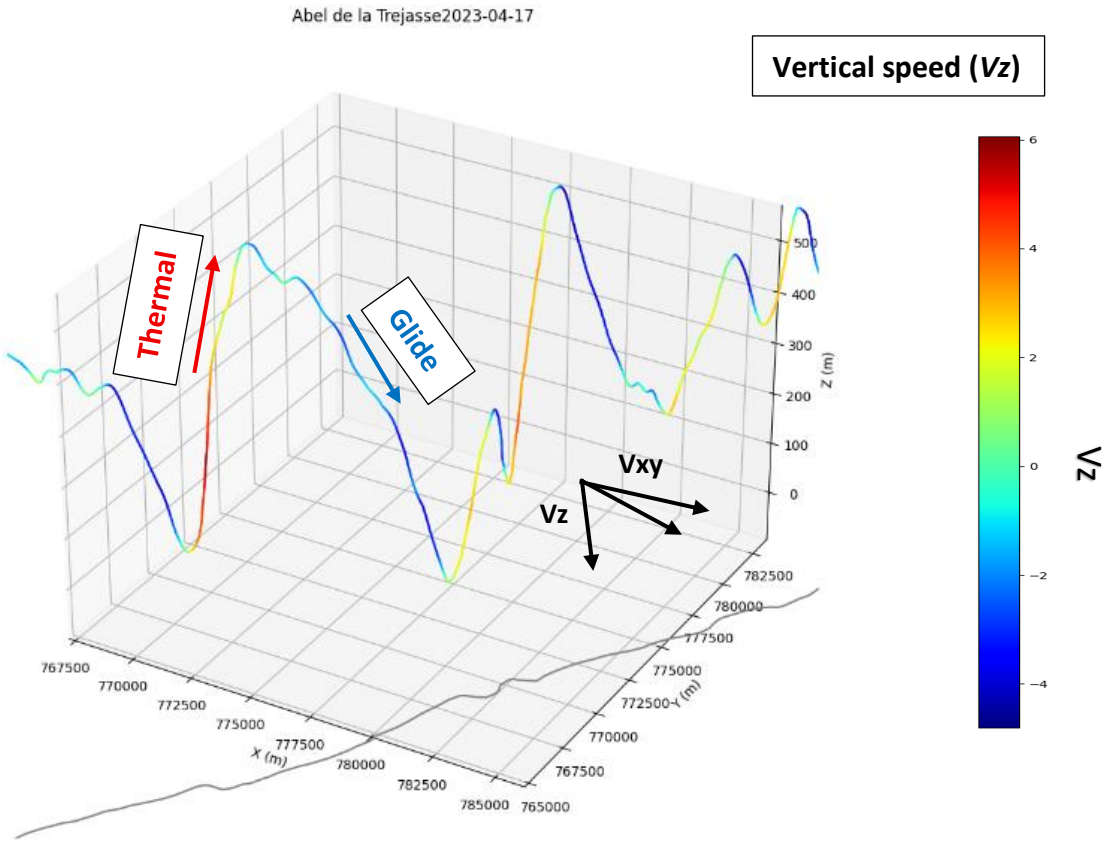


Figure 2. 3D thermal-glide flight sequence, with warm colors indicating positive vertical speed (*Vz*) (thermal) and cool colors indicating negative vertical speed (*Vz*) (glide), and the corresponding 2D trajectory projected in grey. Example from Abel de la Trejasse, an inexperienced Short-toed Snake Eagle from the 2023 dataset, recorded on 17 April 2023.

This entire Cross-Country Optimization Strategy (CCOS) analysis relies on the method developed by Keskin *et al.* (2025). Thermal identification, using high-frequency data only, defined a thermal as a period of at least 36 consecutive seconds (3 full turns) during which the bird circled within a radius of less than 100 m while climbing at a horizontal speed greater than 3 m/s. Unlike the classification approach used in the previous section, this method, performed using Python (version 3.14.2), identifies only "reliable" thermals, for which all speed-related metrics can be calculated and averaged with high precision, thereby restricting the dataset but increasing the accuracy of the resulting estimates. Evaluation of gliding and thermalling phases of flight was based, for each thermalling phase, on the extraction of circling radius, mean horizontal speed, and mean climb rate [Appendix 6 a]. Gliding phases, in which birds move forward while losing height, were identified using the same vertical-speed and curvature thresholds described above, yielding an "empirical" glide polar rather than a theoretical pure-gliding polar (Keskin *et al.* 2025; Pennycuick *et al.* 2012). Airspeed during gliding was estimated by subtracting daily average wind velocity from ground speed [Appendix 6 b, c]. Days with no detected thermals were excluded, since the analysis focused on the relationship between thermalling and gliding behavior; all reported horizontal speeds thereafter refer to estimated airspeed. Following the approach of Keskin *et al.* (2025) empirical glide polar was fitted as a second-order function, $f(x) = ax^2 + bx + c$, to the measured relationship between horizontal speed and sink rate during gliding. From this fit, the maximum glide ratio was calculated as $\sqrt{(c/a)}$, and the airspeed at minimum sink was obtained by converting the quadratic to vertex form (x-coordinate = $-b/2a$). The best-glide airspeed corresponds to the point beyond which the glide ratio (forward distance per unit height lost) stops increasing and begins to decline. Because overall cross-country speed depends on the inter-thermal gliding speed chosen between thermals, we characterized this choice using a linear approximation relating horizontal gliding speed (*vxy*) to climb speed (*v_Thermal climb*), after removing outliers beyond two standard deviations from a Gaussian fit. The fitted line, $vxy = A \cdot v_Thermal\ climb + B$, describes the observed strategy: *A_Observed* reflects how adaptively an individual adjusts its inter-thermal speed to thermal strength ($A = 0$ indicating no adjustment), while *B_Observed* captures the preferred gliding speed under near-zero thermal conditions.

Wind and drift Ratio

Wind data were extracted from the ERA5 reanalysis (“ECMWF” 2026) (0.25° resolution, hourly time step) at the 925 hPa pressure level, corresponding to approximately 750 m altitude, in line with the average flight height of Short-toed Snake Eagles, over the zone 15°N–34°N,

20°W–30°E, for the 2020–2026 period. GPS tracks, subsampled at 5-minute intervals, were aggregated into hourly segments (≥5 successive flight positions). For each hourly segment, a reference direction (theta_reference) was calculated as the rhumb-line bearing between the current position and the exit point, representing the direction the bird would take in the absence of wind. The zonal (*u*) and meridional (*v*) wind components were matched to each hourly movement segment according to its position and timestamp. Wind direction is a key factor shaping migratory flight, as birds must continuously adjust their trajectory to counteract crosswind drift, which would otherwise deviate them from their intended route and increase energy expenditure and travel time. Tailwind refers to wind blowing in the same direction as the bird's flight, while crosswind refers to wind blowing perpendicular to it, pushing the bird sideways off its intended path. Tailwind and crosswind components were calculated following the formulas of Vansteelant *et al.* (2017b). The bird's perpendicular displacement speed was compared to the crosswind component to estimate a drift-compensation index (drift ratio), quantifying the extent to which the bird corrected its trajectory to compensate for crosswind. On this basis, each hourly segment was classified into one of six behavioral response categories to crosswind, following the typology of Vansteelant *et al.* (2017b) : overcompensation (*OC*), full compensation (*FC*), partial compensation (*PC*), full drift (*FD*), overdrift (*OD*), and no significant sidewind (*NSW*, threshold < 5 km/h).

II.4 Statistical Analyses

Linear mixed models (LMMs, *lme4*/*lmerTest*) tested effects of experience, season, and their interaction on route efficiency, flight performance, and thermalling metrics, with individual as a random effect. A significant experience×season interaction would indicate that the effect of experience on flight performance is not constant but instead differs depending on the period of migration, reflecting how experience-related benefits may be enhanced or masked by seasonally varying environmental conditions (e.g., wind support, thermal strength). Significant interactions were interpreted via estimated marginal means (*ggpredict*). Cross-country optimization strategy was assessed by fitting climb speed–gliding speed relationships per experience class, compared using a permutation test (n = 9,999 permutations), with Monte Carlo propagation of uncertainty in the derived glide polar parameters (Keskin *et al.* 2025). Wind drift-compensation behavior (six categories) was tested against experience class and geographical location using chi-square tests, with Cramér's V as effect size.

III. Results

III.1 Flight performances

III.1.1 Migration route efficiency

To assess whether migratory efficiency differed according to experience and season, we compared route tortuosity, distance travelled, and crossing duration during Sahara crossing across experience classes and seasons (Fig. 3).

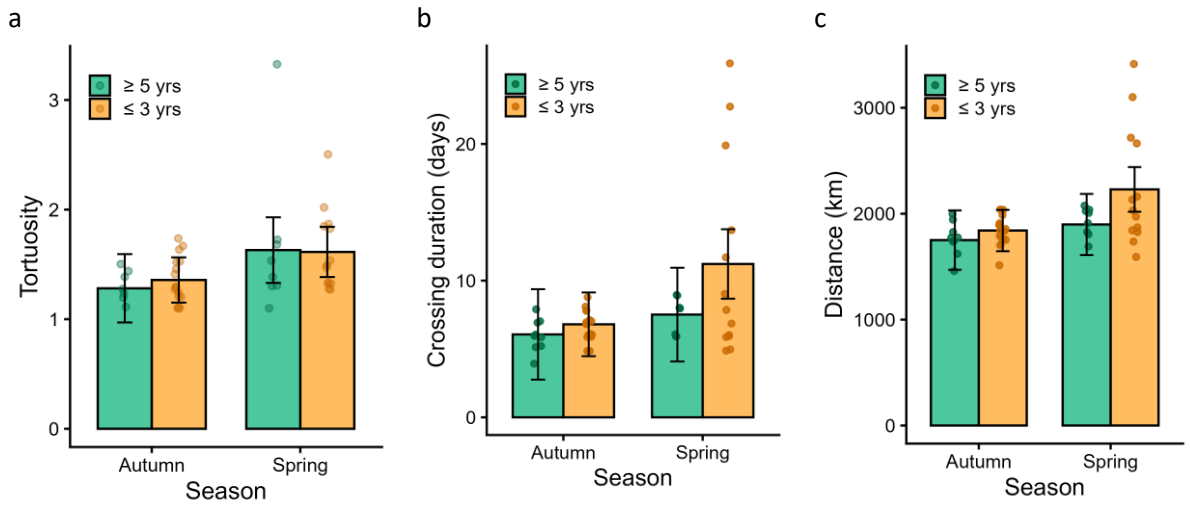


Figure 3. Experience-related variation in migration route efficiency by season. Bars represent estimated marginal means (± 95% CI) from linear mixed models, for tortuosity, distance travelled, and crossing duration, as a function of experience class (≥ 5 yrs, ≤ 3 yrs) and season (autumn, spring) on 46 migrations (2020–2026) of 13 eagles (4 experienced, 9 inexperienced). None of experience×season interactions, nor experience or season, were statistically significant for any of the metrics ($p > 0.05$).

Tortuosity, calculated over the entire Sahara crossing, ranging on average from 1.2 to 1.5, values close to 1, indicating relatively low route tortuosity. The experience:season interaction was not significant ($\beta = -0.093 \pm 0.230$, $p = 0.689$), nor was experience ($\beta = 0.076 \pm 0.185$, $p = 0.685$) or season ($\beta = 0.349 \pm 0.186$, $p = 0.070$), although the latter suggested a trend toward higher tortuosity in spring. The model explained a limited proportion of variance (marginal $R^2 = 0.127$; conditional $R^2 = 0.233$), indicating that inter-individual variability and fixed effects contributed equally to the variance of the data. Sahara crossing distances, measured between the entry and exit points of the critical zone (Fig. 1, spanning 17.5°N to 28°N, roughly 1,165.5 km in a straight north–south line), ranged from 1,444.9 km (experienced) to 1,485.5 km (inexperienced) in autumn, and from 1,615.4 km to 1,763.6 km in spring, staying close to this straight-line distance. The interaction was not significant ($\beta = 241.23 \pm 149.43$, $p = 0.116$), nor was experience ($\beta = 90.20 \pm 169.25$, $p = 0.602$) or season ($\beta = 147.41 \pm 117.45$, $p = 0.218$). Crossing durations ranged from 6.1 ± 3.2 days (experienced) to 6.8 ± 2.2 days (inexperienced) in autumn, and from 7.5 ± 3.5 days (experienced) to 11.2 ± 2.5 days (inexperienced) in spring,

meaning individuals spent close to a week in this resource-scarce barrier. The experience:season interaction was not significant (β = 2.965 ± 2.010, p = 0.149), nor was experience (β = 0.739 ± 2.003, p = 0.717); season was also not significant (β = 1.451 ± 1.583, p = 0.366), though spring crossings were on average longer than autumn crossings.

Despite the lack of statistically significant interaction and experience effects, relative variation values suggest some trends: both distance and duration were higher in spring than in autumn across experience classes, with distance increasing by +11.8% in experienced and +18.7% in inexperienced individuals, and duration increasing by +23.0% in experienced and +64.7% in inexperienced individuals, while tortuosity showed no marked difference between classes.

III.1.2 General flight performance metrics

To further evaluate whether migratory experience shapes flight performance, we compared ground speed, flight height, and altitude gain across experience classes and seasons (Fig. 4).

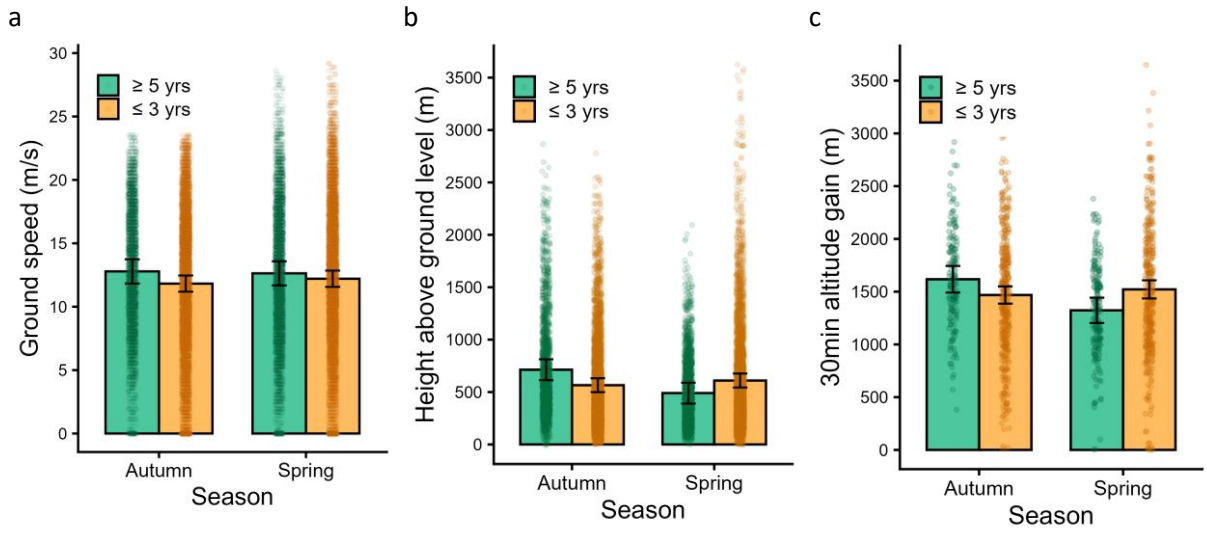


Figure 4. Experience-related variation in general flight performance metrics by season. Bars represent estimated marginal means (± 95% CI) from linear mixed models, for ground speed, height above ground level, and 30-min altitude gain, as a function of experience class (≥ 5 yrs, ≤ 3 yrs) and season. All metrics showed a significant experience×season interaction ($p < 0.005$), with inexperienced individuals underperforming experienced in autumn but matching or exceeding them in spring.

Predicted ground speed (not airspeed, and thus reflecting the influence of wind), based on 21,413 measurements, was slightly lower in inexperienced individuals than in experienced individuals in both seasons, though the gap narrowed from autumn (11.8 vs 12.8 m/s, −7.5%) to spring (12.2 vs 12.6 m/s, −3.4%), consistent with the positive interaction term. These values, equivalent to roughly 42–46 km/h, are broadly consistent with cruising speeds reported for large soaring raptors during active migratory flight. The interaction was significant (β = 0.537 ± 0.169, $p < 0.005$, marginal $R^2 = 0.004$, conditional $R^2 = 0.032$). Flight height (n = 11 297

measures) and altitude gain (n = 1 114 tracks) reflect flight performance in terms of altitude gain during ascending flight. Birds flew at heights ranging from roughly 490 to 713 m above ground, gaining between 1,322 and 1,617 m of altitude over 30 minutes of flight. This pattern reversed between seasons: inexperienced individuals flew lower and gained less altitude than experienced individuals in autumn, but flew higher and gained more altitude than experienced individuals in spring. Indeed, for flight height, the interaction was highly significant ($\beta = 267.39 \pm 18.27$, $p < 0.001$, marginal $R^2 = 0.021$, conditional $R^2 = 0.068$). Predicted flight height was lower in inexperienced than in experienced in autumn (565.5 m vs 713 m, −20.7%), whereas the pattern reversed in spring, with inexperienced flying higher (609.9 m vs 490 m, +24.4%). For altitude gain, the interaction was also significant ($\beta = 348.02 \pm 73.14$, $p < 0.001$, marginal $R^2 = 0.024$, conditional $R^2 = 0.048$). Inexperienced gained less altitude than experienced in autumn (1,468.2 m vs 1,617 m, −9.2%), but more in spring (1,521.6 m vs 1,322 m, +15.0%).

III.1.3 Use of thermal updrafts

Beyond broad performance metrics, we examined thermalling behavior itself, how often birds engaged thermals and how efficiently they exploited them, by comparing thermal frequency, duration, and altitude gain per thermal across experience classes and seasons (Fig. 5).

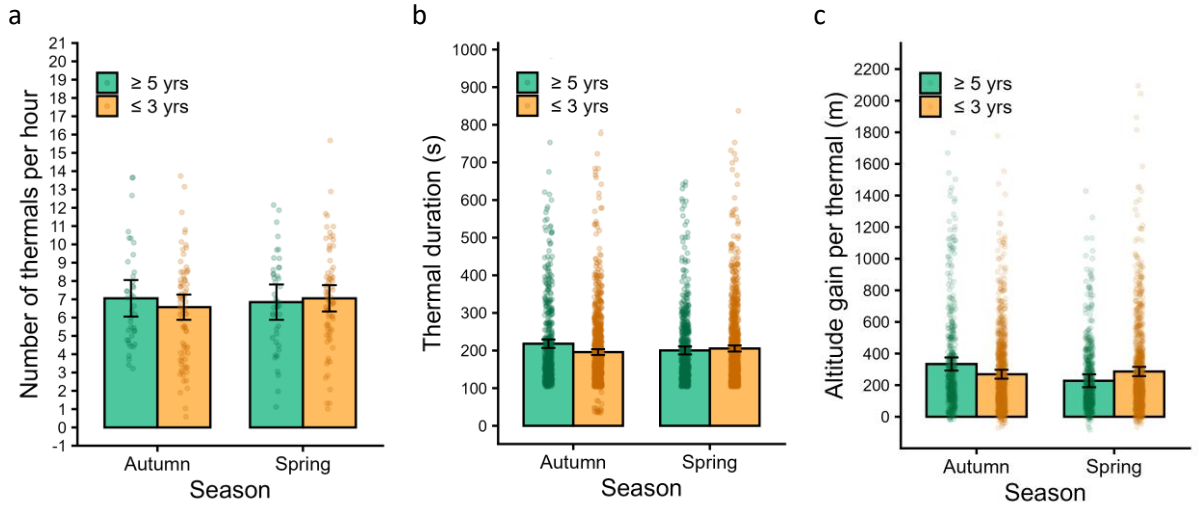


Figure 5. Experience-related variation in thermalling performance by season. Bars represent estimated marginal means (± 95% CI) from linear mixed models, for thermals per hour, altitude gain per thermal, and thermal duration, as a function of experience class (≥ 5 yrs, ≤ 3 yrs) and season (autumn, spring). Thermalling frequency showed no significant interaction, whereas altitude gain and duration both showed inexperienced underperforming in autumn but matching or exceeding experienced in spring.

Thermalling behavior can be characterized by how often birds engage in thermal soaring, and how efficiently they use each thermal in terms of duration and altitude gained.

Thermalling frequency (n = 243 tracks) ranged from approximately 6.6 to 7.1 thermals per hour across seasons and experience classes; combined with ground speed, this corresponds to birds

engaging a new thermal roughly every 6 to 7 km travelled. Neither the interaction (β = 0.693 ± 0.785, p = 0.378) nor the main effects of experience (β = −0.485 ± 0.616, p = 0.438) or season (β = −0.209 ± 0.633, p = 0.741) were significant, and the model explained little variance (marginal R^2 = 0.006, conditional R^2 = 0.026), indicating no clear experience or season-related pattern in thermalling frequency. Thermalling efficiency (n = 3,106 thermals, of which 59 [1.9%] showed a negative altitude gain) was assessed through thermal duration and altitude gain per thermal. Thermal duration ranged from roughly 196 to 218 seconds (about 3.3 to 3.6 minutes) per thermalling event, during which birds gained between 228 and 334 m of altitude. Both variables showed a significant experience×season interaction (duration: log-transformed β = 0.111 ± 0.03, p < 0.001, marginal R^2 = 0.006, conditional R^2 = 0.011; altitude gain: β = 0.429 ± 0.115, p < 0.001, marginal R^2 = 0.014, conditional R^2 = 0.030), following the same reversal observed for flight height and 30-min altitude gain. Back-transformed predicted values were lower in inexperienced individuals than in experienced individuals in autumn (duration: 195.9 s vs 218.1 s, −10.2%; altitude gain: 269.1 m vs 333.6 m, −19.3%), but higher in spring (duration: 205.5 s vs 200.3 s, +2.6%; altitude gain: 286.4 m vs 227.8 m, +25.7%).

III.2 Cross-country optimization strategy

The Cross-Country Optimization Strategy analysis was based on 13 individuals (4 experienced, 9 inexperienced) across 74 migration-days, comprising 3,583 thermal points and 75,731 gliding points, used to derive the glide polar and flight-optimization parameters (Table 1).

Table. 1 : Gliding flight metrics and cross-country optimization strategy parameters, derived from the empirical glide polar, for all individuals. Best glide ratio and horizontal speed at best glide (Vxy) are reported with their associated errors. AObserved and BObserved reflect, respectively, adaptivity to thermal strength and the preferred inter-thermal gliding speed under near-zero thermal conditions.

Best glide ratio	Vxy at best glide (m/s)	Min sink (m/s)	A linear	B linear (m/s)
9.5 +/- 0.2	9.16 +/- 0.81	-0.45 +/- 2.96	0.72 +/- 0.22	13.76 +/- 0.56

The best glide ratio (9.5 ± 0.2) and *Vxy* (9.16 ± 0.81 m/s) indicate the birds cover roughly 9.5 m horizontally per metre of altitude lost at a horizontal speed *Vxy* of 9.16 ± 0.81 m/s. The near-zero minimum sink rate (−0.45 ± 2.96 m/s) likely reflects the shallow polar curvature (Fig. 6) rather than a true estimate, consistent with its margin error. The *B_linear* intercept (13.76 ± 0.56 m/s), the preferred gliding speed under near-zero thermal conditions, close to *Vxy*, indicates birds default to a near-optimal cruising speed under near-zero thermal conditions.

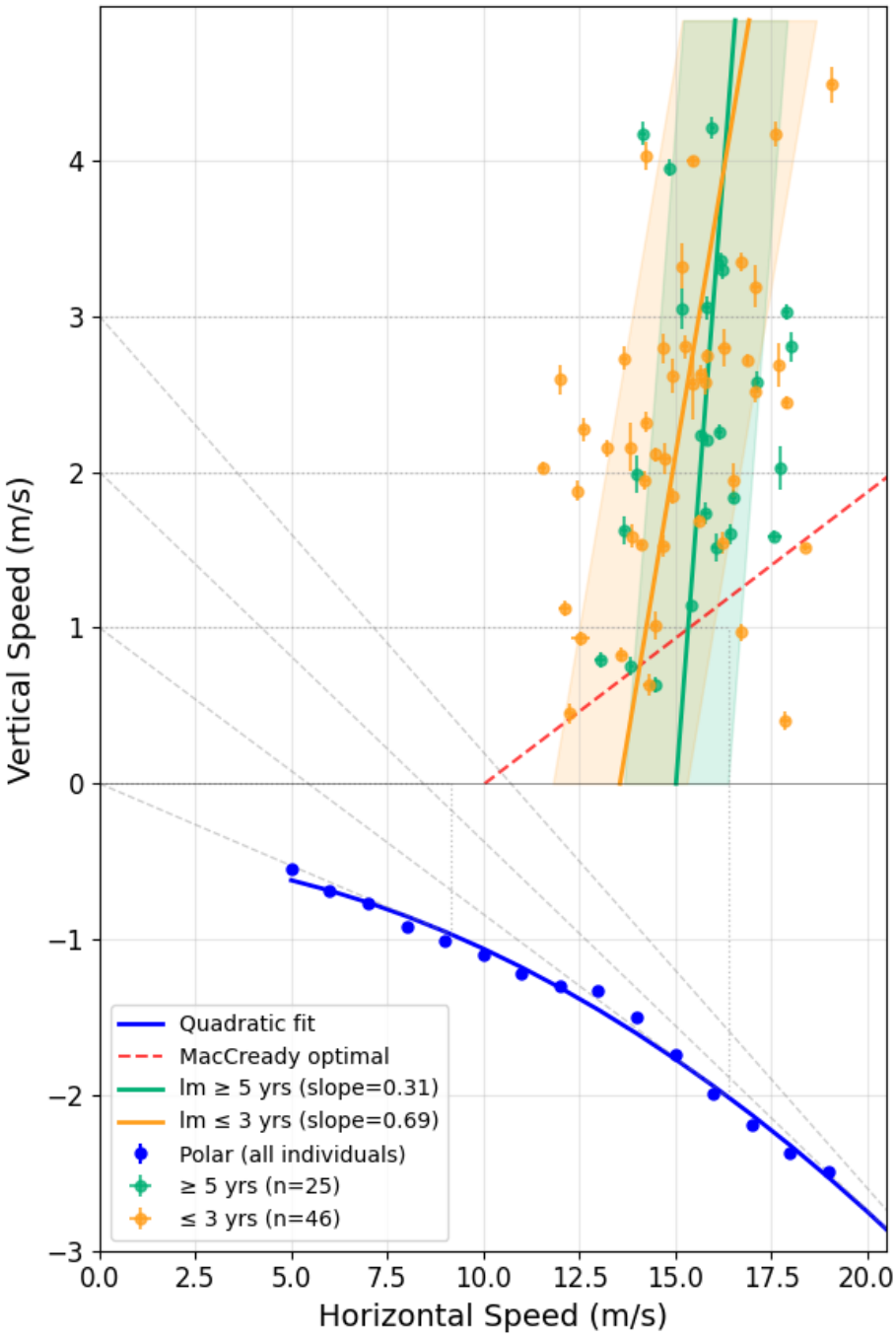


Figure 6. Cross-country optimization strategy and experience-class comparison.
The solid blue line shows the empirical gliding polar curve (lower half), while the dashed red line represents the optimal soaring strategy predicted by MacCready theory (upper half). Colored circles show daily individual means for each experience group (≥ 5 yrs, green and ≤ 3 yrs, orange), with error bars indicating the standard error of the mean; colored lines show linear fits per experience, with shaded bands representing the associated confidence intervals. As in Keskin *et al. (*2025), we used a linear approximation of the optimal soaring curve for easier comparison with empirical data. Vertical speed (*Vz*) is the independent variable in the linear fit: the slope *(A)* reflects sensitivity to thermal strength, while the intercept *(B)* captures the preferred inter-thermal gliding speed under near-zero thermal conditions. Birds showed significant, though partial, adjustment of gliding speed to thermal strength relative to the MacCready optimum, with no significant effect of experience on either slope or intercept (permutation test, n = 9,999).

To assess whether birds optimized their cross-country flight, we compared the observed relationship between climb speed and inter-thermal gliding speed to the theoretical optimum predicted by MacCready theory (which predicts that gliding speed should increase with anticipated thermal strength; Fig. 6). Birds significantly adjusted their inter-thermal gliding speed to thermal strength (*A_Observed* = 0.72 ± 0.22, distinct from zero, Tab. 1), but this adaptation remained comparatively weak relative to the steeper relationship predicted under the MacCready framework (dashed line, Fig. 6a), suggesting that Short-Toed Snake Eagles only partially optimize their cross-country speed to the thermal conditions encountered. To assess whether experience influenced this strategy, we compared the slope and intercept of the climb speed–gliding speed relationship between experience classes, using a permutation test (n = 9,999 permutations) [Appendix 7]. The observed differences in slope and intercept fell within the bulk of their respective null distributions (Fig. 6), indicating no significant effect of experience on either sensitivity to thermal strength (slope) or preferred inter-thermal gliding speed under near-zero thermal conditions (intercept). These results suggest experienced and inexperienced individuals adopt a similarly cross-country strategy. Further analyses suggested the significance of *A_Obs* was driven mainly by inexperienced individuals [Appendix 8].

III.3 Decision-making according to wind conditions

III.3.1 Wind drift/compensation behaviors

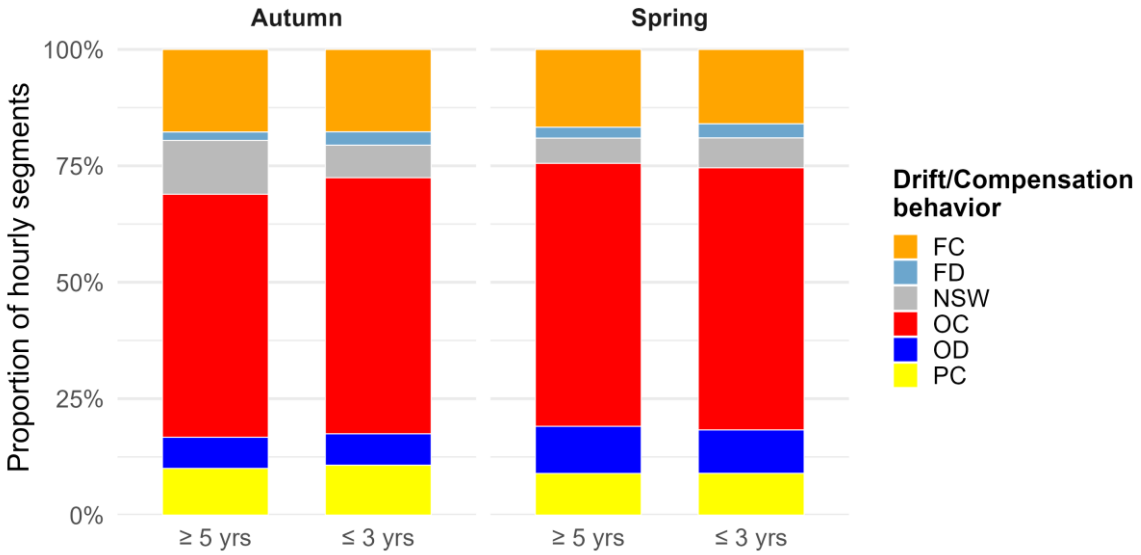


Figure 7. Proportion of hourly segments classified by wind drift-compensation behavior, as a function of experience class (≥ 5 yrs, ≤ 3 yrs) and season (autumn, spring). Categories: overcompensation (OC, red), full compensation (FC, orange), partial compensation (PC, yellow), full drift (FD, light blue), overdrift (OD, dark blue), no significant sidewind (NSW, grey; threshold < 5 km/h). Chi-square tests revealed the Drift/Compensation behaviors did not differ by experience in either season, but varied significantly, though modestly, with season in experienced individuals only.

Wind responses can be drifting with the wind, accepting lateral displacement to save energy, or by compensating for it through active trajectory. To test whether wind compensation strategy varied with experience or season, we computed the proportion of hourly segments (n = 2,471 total; experienced = 814, inexperienced = 1,657) falling into each behavioral category as a function of experience class and season (Fig. 7). The association between experience class and wind drift compensation behavior was then tested separately for each season using chi-square tests [Appendix 9 a]. Neither test revealed a significant association, in either autumn ($\chi^2 = 8.35$, df = 5, p = 0.138, Cramér's V = 0.08) or spring ($\chi^2 = 1.20$, df = 5, p = 0.945, Cramér's V = 0.03), indicating that overall compensation strategies were broadly similar between inexperienced and experienced individuals across both migration seasons. The associated Cramér's V values indicated a negligible effect size in both cases, so these non-significant results should be interpreted with confidence rather than caution regarding statistical power. The association between season and wind drift compensation behavior was tested separately for each experience class using chi-square tests [Appendix 9 b]. The effect of season on wind drift compensation behavior differed between experience classes. Among inexperienced individuals, no significant association was found between season and compensation behavior ($\chi^2 = 5.80$, df = 5, p = 0.326, Cramér's V = 0.06). Among experienced individuals, however, the association was significant ($\chi^2 = 13.40$, df = 5, p = 0.020, Cramér's V = 0.13), although the effect size remained small, indicating that season explains only a modest share of the variation in compensation behavior among experienced individuals.

Wind drift-compensation behavior was largely unaffected by experience, whereas season significantly, though modestly, shaped compensation strategy in experienced individuals only.

III.3.2 Influence of wind patterns on migration route

At this altitude (~750 m, 925 hPa), winds over the Sahara are predominantly easterly, driven by the descending branch of the Hadley cell. Because this represents a persistent crosswind relative to the birds' broadly north–south migratory heading, the resulting drift risk is expected to vary spatially depending on local wind strength and direction along the flyway. To examine whether wind drift-compensation behavior varied geographically along the flyway, we compared the frequency of wind drift/compensation behaviors within each 2° latitude band to the seasonal flyway average, using chi-square tests (Table 2; Fig. 8).

Overcompensation (*OC*) frequencies significantly exceeded the seasonal flyway average in four latitude bands, at opposite ends of the flyway depending on season (Table 2). In autumn, *OC* was significantly elevated in the central portion of the flyway, at 23°–25°N (residual = +15.5%,

$p < 0.001$) and 25°–27°N (residual = +9.9%, $p = 0.008$). In spring, *OC* was instead significantly elevated in the two southernmost bands, at 17°–19°N (residual = +21.8%, $p < 0.001$) and 19°–21°N (residual = +15.7%, $p < 0.001$), corresponding to the early stage of migration during the northbound ascent. Overdrift (*OD*) showed a more limited spatial signal, with a single significant elevation in spring at 23°–25°N (residual = +4.7%, $p = 0.010$).

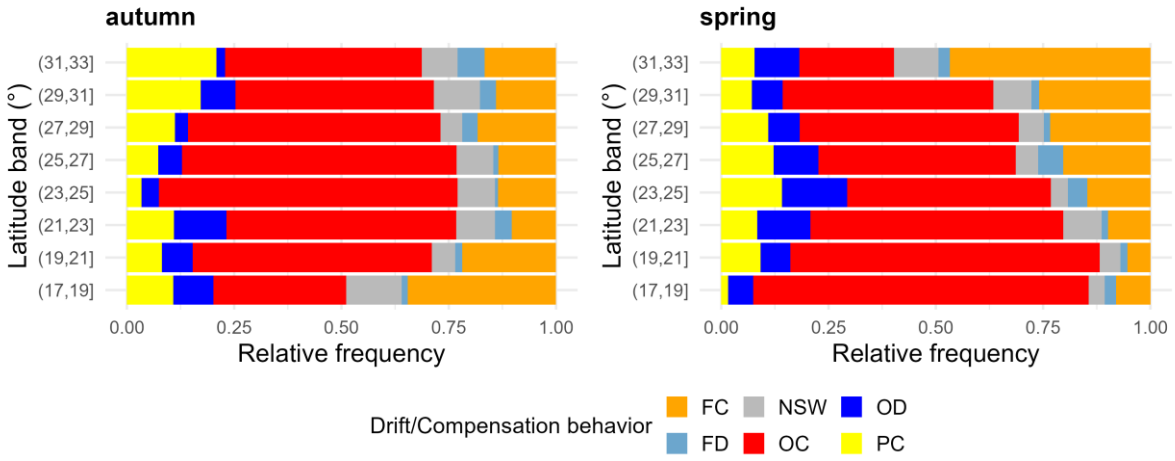


Figure 8. Geographical flexibility in hourly orientation strategies and wind conditions across the flyway during autumn and spring migration. Categories: overcompensation (OC, red), full compensation (FC, orange), partial compensation (PC, yellow), full drift (FD, light blue), overdrift (OD, dark blue), no significant sidewind (NSW, grey; threshold < 5 km/h). Compensation was significantly higher at 17–19°N in spring and at 23–25°N in autumn.

Table 2. Residual frequencies of overcompensation (OC) and overdrift (OD) in comparison with the remaining orientation behaviors ('Other') across each 2° latitude band during autumn and spring migration. Residual frequencies were computed relative to the average frequency of each behavioral class across the entire flyway for each season, using a chi-square test. Significantly higher frequencies of OC and OD than the seasonal flyway averages are highlighted in bold.

Lat (°)	Autumn			Spring		
	OC	Other	OD	OC	Other	OD
(31,33]	-8.3	12.9	-4.6	-34.3	33.5	0.8
(29,31]	-7.9	6.5	1.3	-7.2	9.7	-2.4
(27,29]	4.8	-1.1	-3.7	-5.3	7.6	-2.3
(25,27]	**9.9**	-8.7	-1.2	-10.4	9.6	0.9
(23,25]	**15.5**	-12.8	-2.7	-8.9	3.2	**5.7**
(21,23]	-0.5	-5.0	**5.5**	2.6	-5.3	2.8
(19,21]	1.6	-2.0	0.4	**15.7**	-13.1	-2.6
(17,19]	-23.2	20.5	2.6	**21.8**	-18.1	-3.7

IV. Discussion

We hypothesized that experience would significantly influence flight performance, and that season would also affect flight behavior, potentially interacting with experience, with more challenging conditions expected in spring, particularly for experienced (breeding) individuals. The first hypothesis was not supported: experience alone had little influence across nearly all variables considered. The second was largely confirmed, though not uniformly. Season, and its interaction with experience, had a marked effect on flight height, altitude gain, and thermalling performance, but no supported effect on migration route efficiency (tortuosity, distance, crossing duration), despite a trend toward longer crossing distance and duration in spring. Migration route efficiency did not differ between experienced and inexperienced individuals, nor across seasons. Flight height, altitude gain, speed and thermalling performance, by contrast, showed a marked experience×season interaction, with inexperienced individuals underperforming compared to experienced individuals in autumn but matching or exceeding them in spring. A reversal that, rather than reflecting a simple improvement with experience, largely coincides with the fact that experienced individuals migrate two months earlier than inexperienced individuals in spring, when wind and thermal conditions are less favorable for migrating. Cross-country flight optimization revealed that birds do adjust their inter-thermal gliding speed to thermal strength, but only weakly relative to the optimum predicted by the MacCready theory. Again, this adjustment did not significantly change with experience. Similarly, wind drift-compensation behavior showed no significant association with experience, although it varied systematically with geographic position and season, tracking the underlying wind regime encountered along the flyway.

IV.1 An innate or rapidly acquired decision-making process

IV.1.1 A moderate but early-mastered optimization of thermal use

Thermalling occurred at a fairly regular rate across individuals, with birds initiating a thermal roughly six times per hour on average. Thermal soaring accounted for just over half of the total altitude gained during the Sahara crossing, indicating that, while thermalling is clearly the dominant climbing strategy, a substantial proportion of altitude gain is achieved through other means. The Short-toed Snake Eagle thus appears able to exploit other types of updrafts as well, a flexibility also reported in Golden Eagles, which rely heavily on orographic lift over slopes and ridges, in contrast to Turkey Vultures (*Cathartes aura*), which use thermal lift almost exclusively (Bohrer *et al.* 2012). The thermalling frequency did not differed significantly between experience classes or seasons, suggesting that the decision to enter a thermal is largely

opportunistic, triggered by the presence of a usable updraft rather than modulated by prior experience. By contrast, inter-thermal gliding speed was significantly, but moderately, adjusted based on the strength of the current thermal, presumably used by the birds as a proxy for the strength of the next thermal, consistent with the MacCready cross-country optimization strategy. This adjustment was of similar magnitude between experienced and inexperienced individuals, indicating that the ability to anticipate thermal strength and adapt gliding speed accordingly is already present from the very first migration, rather than being progressively refined with experience unlike in Eurasian Griffon Vultures, with adults outperforming juveniles in thermals (Harel *et al.* 2016).

Compared with other raptors, the Short-toed Snake Eagle shows a comparatively moderate adjustment of gliding speed to thermal strength, weaker than that reported for species with lower wing loading, such as the Lesser Kestrel (*Falco naumanni*) (1.6 kg/ m²) (Ákos *et al.* 2008; Keskin *et al.* 2025; Taylor *et al.* 2016). Keskin *et al.* (2025) show that thermal-strength adaptivity scales inversely with wing loading. Species with lower wing loading gain greater benefit from thermalling and can select their inter-thermal speed more boldly than species with higher wing loading (Pennycuick *et al.* 2012; Rayner 1990; Waldrop *et al.* 2020). With an estimated wing loading of around 3.4 kg/m² (based on a mean body mass of 1.8 kg and mean wingspan of 1.85 m), below both the Peregrine Falcon (*Falco peregrinus*, ~6 kg/m²) and the Steppe Eagle (*Aquila nipalensis*, ~4 kg/m²), the Short-toed Snake Eagle would be expected to show comparatively higher thermal-strength adaptivity than either species, but this is not the case. Thermal-strength adaptivity may also depend on factors other than wing loading, for example, greater adaptivity in *F. peregrinus* may partly reflect the manoeuvrability needed to capture aerial prey (Keskin *et al.* 2025). More broadly, closely following MacCready-optimal speeds risks being caught without a usable thermal, forcing a costly switch to flapping flight, so high thermal-strength adaptivity can be seen as a comparatively risk-prone strategy (Richardson *et al.* 2018; Shepard *et al.* 2016). Notably, the flights used for comparison in these other species were not recorded in the context of migration across a critical barrier, under which the Short-toed Snake Eagle's more moderate adaptivity may instead reflect a risk-averse strategy specific to the Sahara crossing.

IV.1.2 A strongly compensatory species, with this ability present from the outset

Migratory journeys spanned broadly similar durations and distances, and followed comparably direct routes, regardless of experience. Indeed, several aspects of the migratory behavior of birds, notably timing, direction and duration, have been shown, mainly from breeding

experiments, to be under genetic influence (Newton 2024). The relative proportions of the different wind drift-compensation behaviors (full compensation, partial compensation, overcompensation, drift, overdrift, and no significant sidewind) were similar between experienced and inexperienced individuals, with no significant association between experience class and compensation category, with a majority of compensation. The predominance of compensatory behaviors appears to be a species-level characteristic. Short-toed Snake Eagles have been described as capable of undertaking particularly large detours as a compensatory behavior to avoid extensive water crossings (Panuccio *et al.* 2013), in contrast to Honey Buzzards, which make greater use of supportive winds, with a higher proportion of drift (Vansteelant *et al.* 2017b). As with thermal soaring behavior (IV.1.1), this indicates that the ability to respond to sidewind, whether by compensating or allowing drift, is already in place from the firsts migrations, with experienced individuals showing no greater tendency to adopt drift, a strategy that could prove particularly risky (Williams *et al.* 2018) over the Sahara for Short-Toed Snake Eagles.

IV.2 Season, as the main source of apparent performance differences

IV.2.1 Breeding pressure in adults and shifting within-season conditions

General flight performance and thermal updraft strength closely tracked the conditions prevailing at different points within a given migration season, rather than differing consistently between seasons as a whole [Appendix 10.1]. Tailwind support was highest at the beginning of autumn, with favourable wind speeds following the same seasonal pattern, conversely favourable wind speeds increased throughout spring [Appendix 10.2]. Furthermore, sand and dust storm activity over the Sahara shifts progressively northward from winter to summer, so southern crossings in February may be more exposed than those in April (Saleh *et al.* 2025). Critically, this seasonal shift in conditions runs in opposite directions between the two migration periods: conditions are most favourable early in autumn and deteriorate as the season progresses, whereas in spring the reverse holds, with conditions improving as the season advances. Because inexperienced individuals consistently migrate later than experienced ones in both seasons, especially in spring [Appendix 10.1], they systematically miss the favourable early-autumn conditions but instead benefit from the improving late-spring conditions, just like in Golden Eagles (*Aquila chrysaetos*), with adults migrating earlier under more variable weather conditions because of breeding pressure, while younger birds make choices to preserve energy (Duerr *et al.* 2015). All else being equal in terms of underlying flight ability, this timing mismatch alone is sufficient to generate an apparent inversion in performance between

experience classes across seasons. Across all flight performance metrics, the same pattern emerged: inexperienced individuals performed below experienced-individual levels during autumn migration, but matched or exceeded experienced performance during spring migration. The same seasonal reversal was found for thermalling behavior: inexperienced individuals showed slightly lower thermal efficiency than experienced individuals in autumn, but matched or exceeded experienced-individual performance in spring, with the exception of thermalling frequency, which showed no significant experience- or season-related variation. Indeed, age-dependent migration strategies often occur *in tandem* with differences in the timing of breeding (Cristol *et al.* 1999; Newton 2024; Zenzal & Moore 2016). Taken together, these results indicate that in Short-Toed Snake Eagle, inexperienced individuals achieve performance remarkably comparable to that of experienced birds, both in flight metrics and in wind-related behaviors across the Sahara crossing, once the confounding effect of season-driven timing differences is accounted for. This contrasts with species where flight efficiency develops more gradually with experience, such as the White Stork, in which juveniles rely more heavily on costly flapping flight than adults and only improve their flight efficiency over the course of the migration itself (Rotics *et al.* 2016). Or, similarly, the marked effect of experience observed in adult Marsh Harriers (Klaassen *et al.* 2011), and even what was expected for Short-toed Snake Eagles themselves (Vidal-Mateo *et al.* 2016).

IV.2.2 Spatial structure of compensation: coastal proximity

Beyond this progressive improvement or deterioration of conditions within a season, the season itself also shapes flight performance through qualitatively different atmospheric structures over the Sahara. In autumn, the wind field is fairly homogeneous across the desert, dominated by relatively uniform north-easterly trade winds. In spring, by contrast, the wind field is less homogeneous, with strong winds concentrated along the western coast and oriented from south to west and a very high drift potential (Shi *et al.* 2022). As a result, birds tend to follow a westward arc in spring, tracking this coastal tailwind. This seasonal contrast in the spatial structure of the wind field means that the conditions encountered along the flyway differ not only in strength but in geographical pattern between the two migration periods, as shown for adult European Honey Buzzards along the east atlantic flyway (Vansteelant *et al.* 2017b). 'Adaptive drift' (Klaassen *et al.* 2011; Vidal-Mateo *et al.* 2016) that is, they mainly tolerate drift at the onset of autumn and spring migration, while they compensate or overcompensate for side winds while approaching their destination. But migrating birds encounter more complex wind fields than the scenarios from which the predictions of 'adaptive drift' were

originally derived, so it is not surprising that they also demonstrate a higher degree of flexibility in orientation behavior than expected from these models, as shown for Marsh Harriers (Klaassen *et al.* 2011) and for the Honey Buzzards (Vansteelant *et al.* 2021). More generally, wind drift-compensation behavior appears to be driven primarily by proximity to the Atlantic coastline: uncorrected drift is most costly where it risks carrying birds out over the ocean, where thermal soaring is impossible and re-orientation options are limited, so compensation were expected to be strongest where the flyway runs closest to the coast and more relaxed further inland, where drift can still be corrected later in the crossing, to avoid being blown out to sea (Klaassen *et al.* 2011). This coastal-proximity logic, combined with the seasonal shift in wind regimes described above, accounts for the observed latitudinal shift in the zone of strongest compensation. In autumn, the flyway runs closest to the Atlantic coast over the central Sahara, so this is where overcompensation peaks. In spring, by contrast, the northward progression of the ITCZ, brings the flyway closest to the coast further south, near the Sahelian margin, early in the northbound crossing, which is where overcompensation is instead concentrated (Shi *et al.* 2022). Short-toed Snake Eagles are indeed able to compensate for the effects of crosswinds to avoid drifting when close to the coast, consistent with this species showing a particularly strong aversion to crossing water: in the area facing the Strait of Messina, it was the only species to show substantial reverse migration approching water barrier (Agostini *et al.* 2016; Panuccio *et al.* 2012). Also, dust storm activity peaks between 17° and 20°N, matching the latitudinal band where spring wind compensation is highest, consistent with birds adjusting more strongly to wind in this dust-storm hotspot (Saleh *et al.* 2025).

IV.3 Migration route choice and experience

IV.3.1 Decoupling of flight performance from route choice

Despite the experience×season differences observed in flight performance, migration routes themselves were broadly similar between experienced and inexperienced individuals, within a given season. This apparent decoupling between performance and route choice is informative in itself. If inexperienced individuals were simply less capable of finding an efficient path through the Sahara, tortuosity would be expected to differ between experience classes, with inexperienced birds taking less direct or more erratic routes, as shown for Red Kite (*Milvus milvus*) with migration routes becaming more direct with age (Chan *et al.* 2024). Instead, similar tortuosity between groups suggests that route selection is not primarily a matter of directness, but rather of choosing a route well suited to the prevailing wind conditions, even if this means following a longer or more curved path (IV.2) (Agostini *et al.* 2017). A comparable pattern has

been reported in the Montagu's Harrier and the Lesser Kestrel, for which winds were shown to be the strongest influence on migration journeys (Limiñana *et al.* 2013).

IV.3.2 Emergence of route fidelity across successive migrations

Route consistency can offer an alternative explanation for the individual variability in route choice observed independently. Migration route repeatability has been documented in the Hoopoe (*Upupa epops*) (van Wijk *et al.* 2016), and the Grey-faced Buzzards (*Butastur indicus*) (Shiu *et al.* 2006). Route consistency across successive migrations also seems to increase with experience, indeed three experienced birds with at least two recorded migrations show higher consistency than inexperienced birds, although the limited sample size currently precludes any formal analysis. Such a pattern would fit a scenario in which the first few migrations involve a phase of exploration, during which several candidate routes may be tested, after which a route becomes established and is then largely repeated, with wind compensation serving to keep the bird on this familiar path rather than to actively re-optimise it each year, as suggested for Bonelli's eagles *Aquila fasciata* (López-López *et al.* 2014).

IV.4 A largely fixed strategy in a solitary long-distance migrant

Taken together with IV.1, IV.2 and IV.3, except for the route consistency hypothesis, results suggest that, for Short-toed Snake Eagles, the core behavioral and decision-making skills required for a successful Sahara crossing, including the ability to detect and respond appropriately to wind conditions, are largely in place from the first migrations and are not markedly refined with subsequent experience. As reflected with the fact that the 9 inexperienced tagged succeeded in crossing the Sahara Desert for their first migration. This contrasts with species less well adapted to solitary migration, such as the Egyptian Vulture *Neophron percnopterus*, for which population decline forces an increasing proportion of inexperienced individuals to migrate without conspecific guidance, leading to sub-optimal route choice and high first-year mortality (monthly survival probability 0.75), including fatal sea crossings of the Mediterranean (Buechley *et al.* 2021; Oppel *et al.* 2015). The Short-toed Snake Eagle, although already highly efficient during its first migrations, does not show marked improvement with experience, a pattern that may reflect a risk-averse strategy in the Sahara (Alerstam & Lindström 1990), as suggested by both its strong compensation for wind, to avoid water barrier (Agostini *et al.* 2016; Panuccio *et al.* 2012), and its comparatively moderate optimization of thermal use to avoid risks (Richardson *et al.* 2018; Shepard *et al.* 2016).

IV.5 Limitations and perspectives

The main limitation of this study is the relatively limited number of tracked individuals and migrations, with an imbalance between experienced and inexperienced individuals (the latter overrepresented). Larger sample sizes would allow the very first crossing to be isolated as a distinct category. This sample size is, however, partly the counterpart of a deliberate choice: restricting the dataset to migration alone rather than pooling it with other phases of the annual cycle, as in broader-scope studies such as Keskin *et al.* (2025). This limits statistical power but strengthens the interpretability of the behavioral signal, since metrics are not diluted by non-migratory movements. High-frequency GPS recording (one point every 3 seconds) is adequate for studying flight performance and wind-response behavior, but proves less sufficient for finer analyses such as the cross-country section, where it reduces the proportion of usable flight segments. Increasing recording frequency in specific areas would improve estimation of flight parameters. Beyond these methodological refinements, such improved data would open the way to investigating the consequences of climate change-induced disturbance in wind patterns and thermal updrafts on the migration of soaring birds (Nourani *et al.* 2017; Yuni *et al.* 2020), as well as exploring the use of other types of updrafts (Bohrer *et al.* 2012).

Conclusion

Overall, this study suggests that the Short-toed Snake Eagle's ability to navigate the Sahara crossing does not depend on experience gradually built up over successive migrations, but is instead largely present from the outset. Season, acting as a proxy for the atmospheric conditions encountered, appears to be a far stronger driver of flight performance than experience itself, although this seasonal effect is itself partly indirect, since only experienced individuals are old enough to be subject to breeding-timing constraints in spring. This pattern is consistent with a solitary migratory strategy in which individuals cannot rely on social learning from conspecifics, and instead appear to draw on an intrinsic, rapidly available capacity to respond appropriately to the challenges of desert crossing from their very first journey. The species also shows only moderate adaptivity to thermal strength, possibly reflecting a risk-averse strategy that limits exposure to the loss of lift during this energetically demanding crossing. Similarly, its strong and largely intrinsic tendency to compensate for wind drift may stem from a limited capacity to sustain flight over water, avoiding drift toward the Atlantic coast, both patterns reflecting the trade-offs imposed by the Sahara and likely contributing to low crossing mortality. Experience may still play a role beyond flight performance, with a preliminary trend toward greater route fidelity across successive migrations in experienced individuals.

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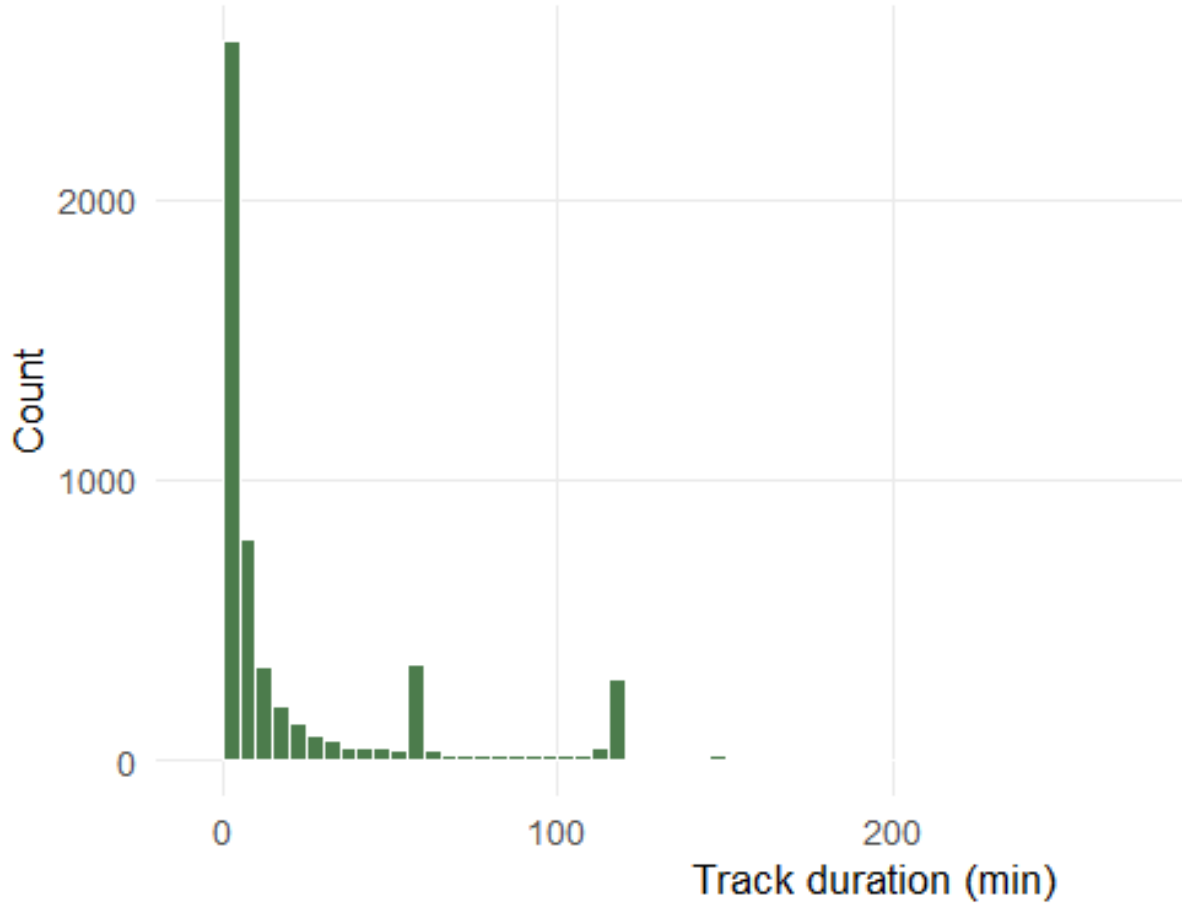
Appendix 10: *Migration periods and comparison of wind conditions……8*

Appendix 10.1: *Migration periods and soaring conditions……8*

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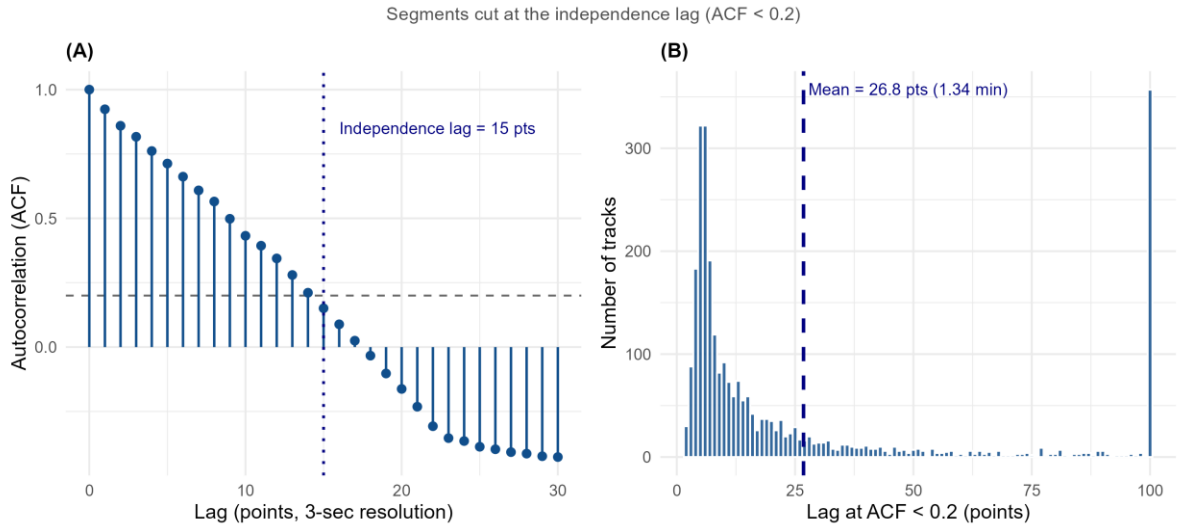
Appendices

Appendix 1 : *Distribution of high-frequency GPS track duration*

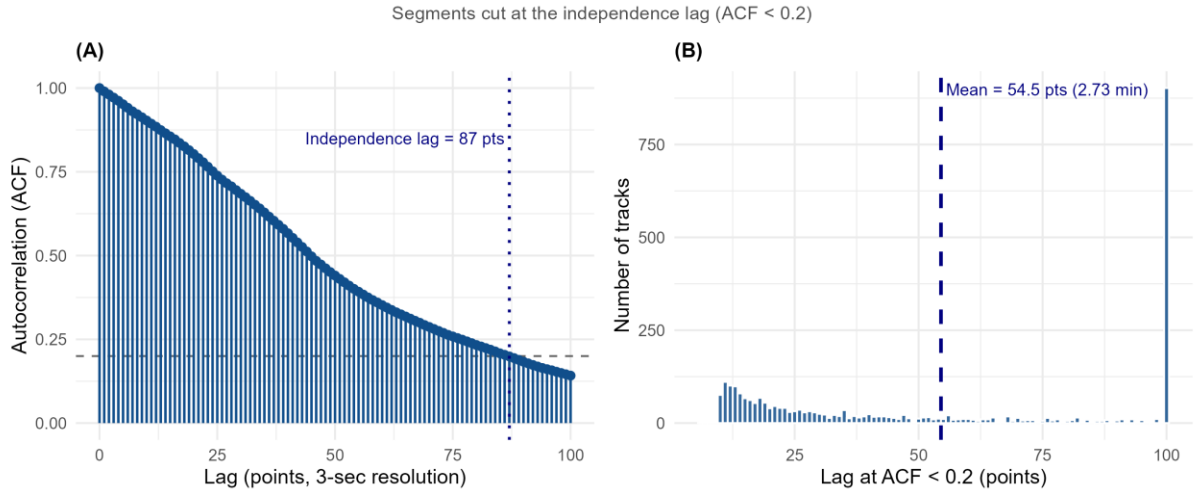


Distribution of high-frequency GPS track duration. Distribution of the duration of contiguous high-frequency GPS tracks (no gap >6 s) used in the analysis. Most tracks were short, with a majority under 10 minutes, alongside secondary peaks around 60 and 120 minutes, likely corresponding to fixed GPS sampling schedules.

Appendix 2 : *Pseudoreplication for ground speed and height above ground*



Determining segment length from ground-speed autocorrelation. (A) Autocorrelation decay (example track). (B) Independence lag across all tracks (= retained segment length). For 357 of 2,846 tracks (12.5%), ground-speed autocorrelation remained above the 0.2 threshold across the entire tested lag range (100 points / 5 min) and was conservatively capped at this value. Including these capped tracks, the mean was 1.34 min (originally 1.51 min prior to ACF lag-range correction).

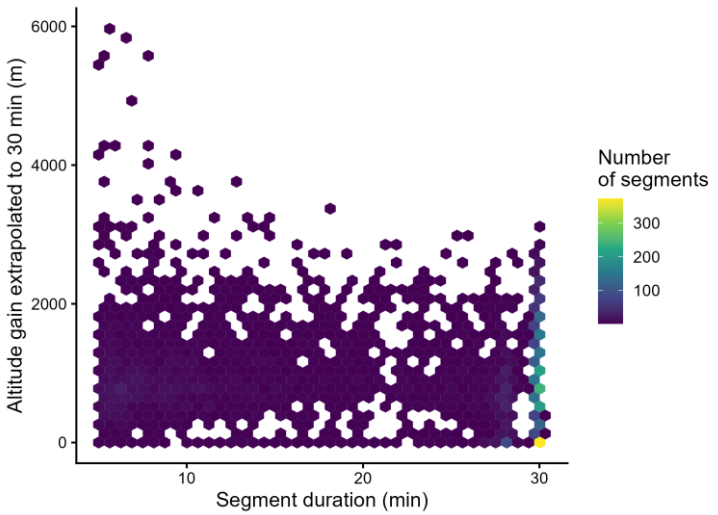


Determining segment length from height-above-ground autocorrelation. (A) Autocorrelation decay (example track). (B) Independence lag across all tracks (= retained segment length). For 901 of 2,846 tracks (31.7%), height-above-ground autocorrelation remained above the 0.2 threshold across the entire tested lag range (100 points / 5 min) and was conservatively capped at this value. Including these capped tracks, the mean was 2.73 min.

Appendix 3 : *30-minute altitude gain calculation*

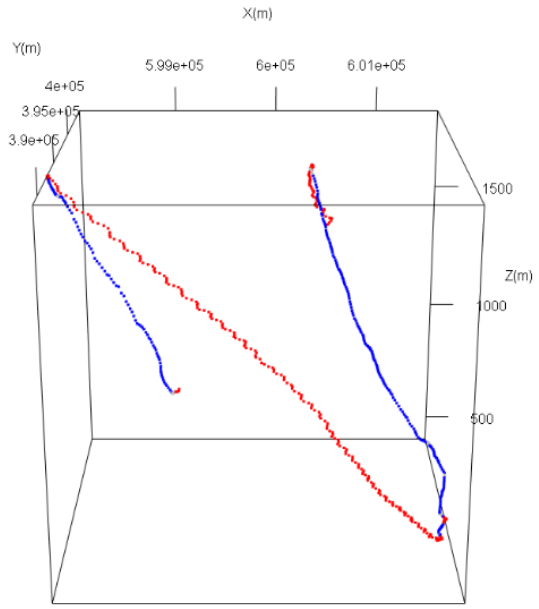
Reference	Duration range tested	n tracks	Error range (%)
1h	1–30 min	11–479	69 – 111
30 min (short→long)	1–30 min	11–479	−4 – 20
30 min (long→short)	40–180 min	11–184	−46 – 22

Extrapolation error by reference duration for standardising altitude gain. Three extrapolation references were tested to determine the most reliable duration for standardising altitude gain: 1 hour, and 30 minutes (tested both upward, from shorter tracks, and downward, from longer tracks). Errors were substantially larger and more inconsistent when extrapolating to 1 hour (69–111%) than when using 30 minutes as the reference (within ±20% upward; up to 46% downward for durations ≥60 min). A 30-minute reference was therefore retained, and tracks exceeding 30 minutes were segmented rather than scaled down, to avoid the larger errors associated with long-duration extrapolation.



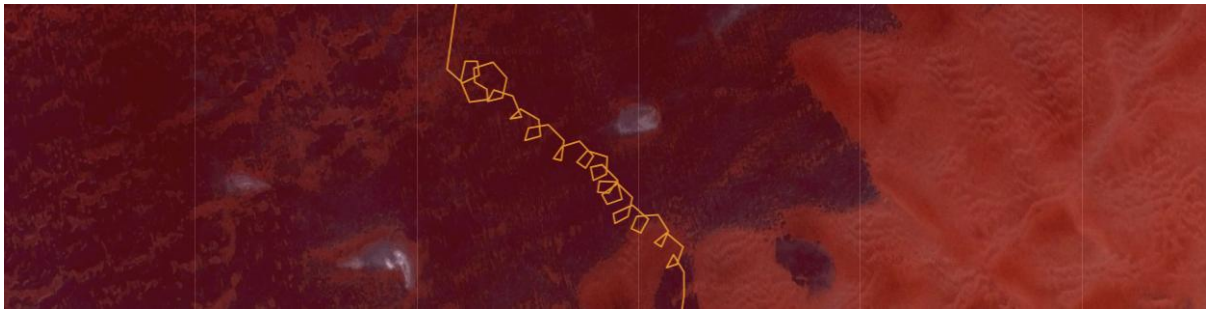
Relationship between extrapolated 30-minute altitude gain and segment duration (high-frequency tracks only). Diamond colour reflects the density of overlapping segments, with warmer colours indicating higher density. Extrapolated 30-minute altitude gain was overestimated for short segment durations ; segments shorter than 5 minutes were therefore excluded from the analysis.

Appendix 4 : *Flight type algorithm*



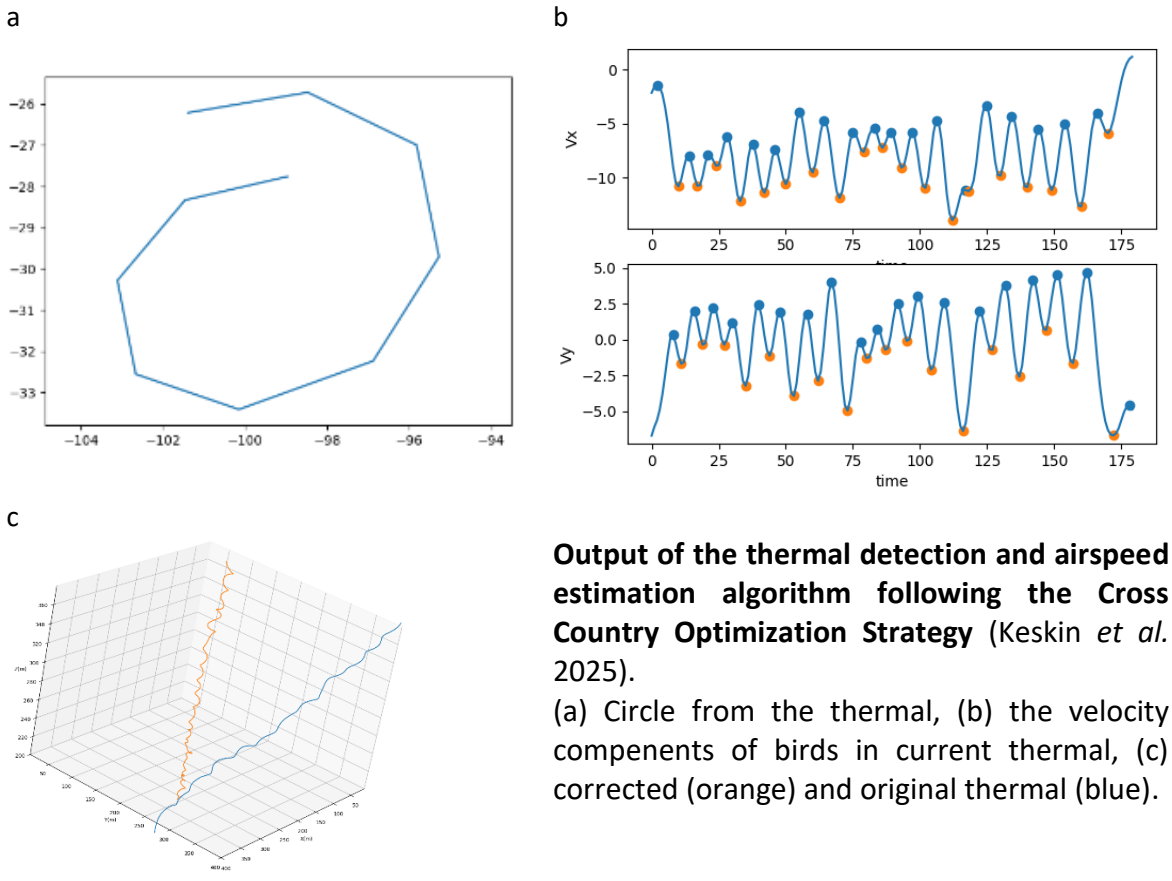
3D representation of the flight type recognition algorithm output. Used to visually check classification accuracy. Thermalling phases ($|K| > 0.01$ m^{-1} and vertical speed $vz > 0.3$ m/s) are shown in red and gliding phases ($|K| < 0.01$ m^{-1}) in blue.

Appendix 5 : *Manual thermal count*



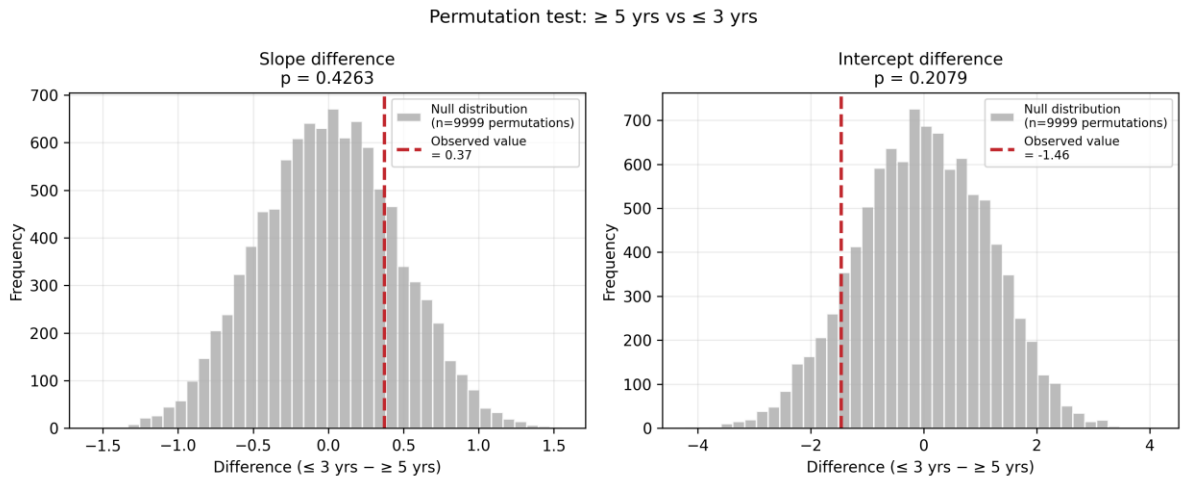
High-frequency GPS track showing a thermalling event (yellow) over the Sahara (Ornitela). Thermal counts were independently assessed from flight trajectories over 6 segments of 1 h and 13 segments of 2 h, and compared with the automated algorithm. Manual counts showed very high agreement between observers, with a correlation of r = 0.98 for both observation durations. For the 1-h segments, the mean number of thermals was 6.5 per segment for both observers, with differences ranging from 0 to 2 thermals. For the 2-h segments, mean counts were 10.38 and 9.13 thermals per segment, respectively, with differences generally limited to 0–2 thermals. This indicates a high consistency in the manual identification of thermals from trajectories. The automated algorithm was tested with and without a 130 m minimum altitude threshold. The unfiltered version, which does not impose a minimum altitude for thermal gain, generally produced higher counts. The version including the 130 m threshold gave more balanced results relative to the manual counts. For the 2-h segments with unambiguous paired manual counts (n = 9), the automated estimate differed from the mean manual count by −0.72 thermals per segment on average, corresponding to a 7.6% lower count. The mean absolute difference was 2.17 thermals, with individual differences ranging from −5 to +4 thermals. Thus, overestimation and underestimation occurred in both directions and partly compensated at the overall level.

Appendix 6: *Cross Country Optimization Strategy algorithm*



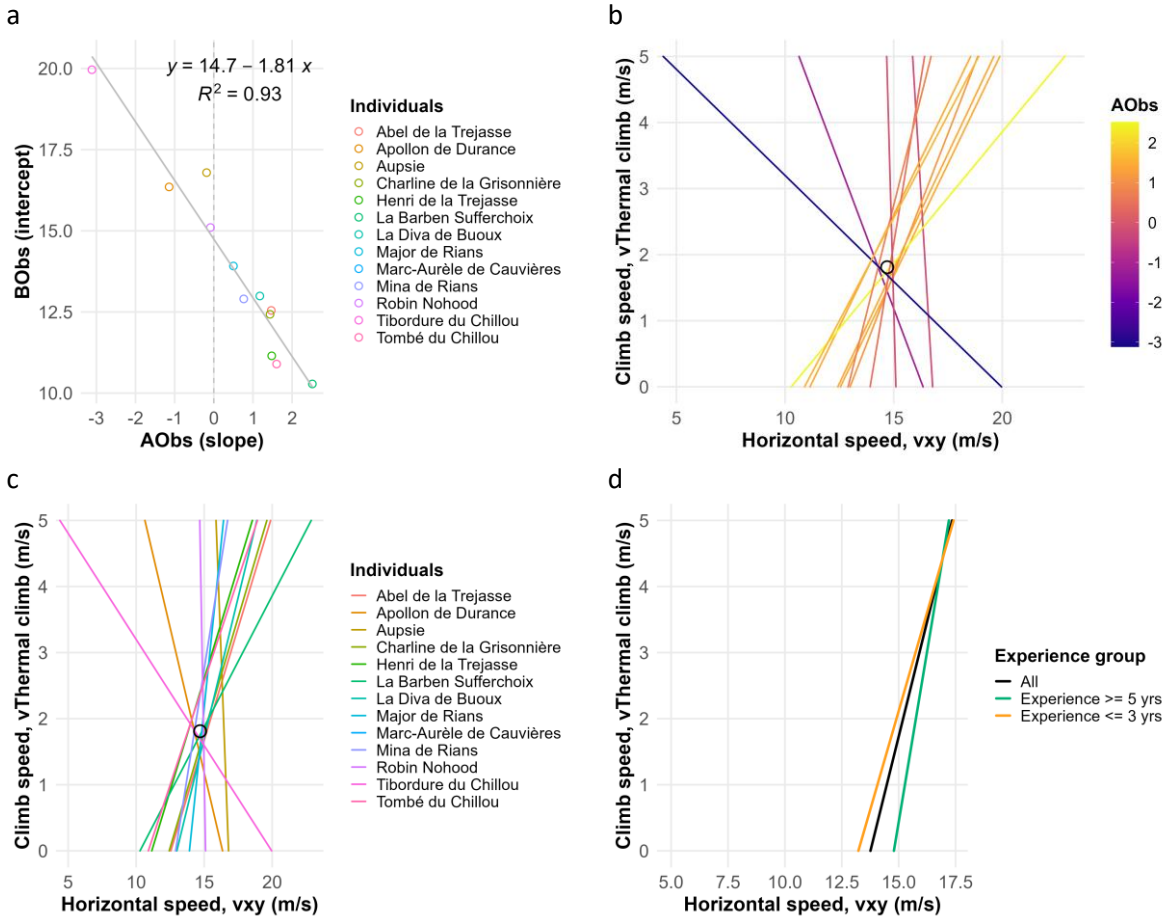
Output of the thermal detection and airspeed estimation algorithm following the Cross Country Optimization Strategy (Keskin *et al.* 2025).
(a) Circle from the thermal, (b) the velocity compenents of birds in current thermal, (c) corrected (orange) and original thermal (blue).

Appendix 7 : *Permutation test Cross Country Optimization Strategy)*



Permutation test comparing the slope and intercept of the climb speed–horizontal speed relationship between experienced and inexperienced. Grey histograms show the null distributions obtained by randomly reshuffling age-class labels across daily observations (n = 9999 permutations); dashed red lines indicate the observed differences. The observed values fall within the bulk of their respective null distributions, confirming that neither the slope nor the intercept differs significantly between age classes.

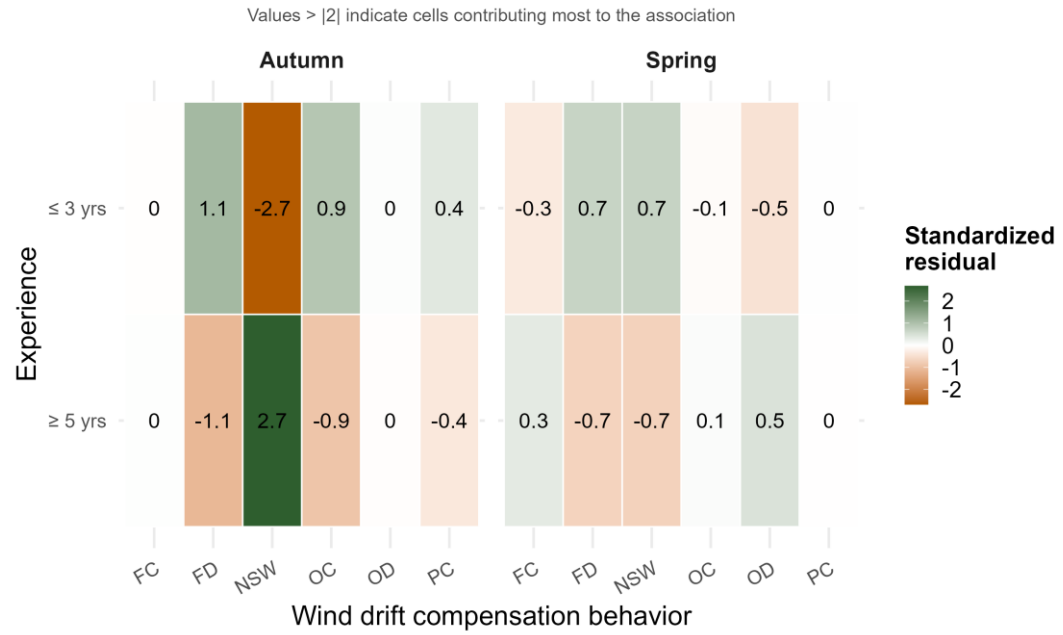
Appendix 8 : *Parameters of Cross-Country Optimization Strategy*



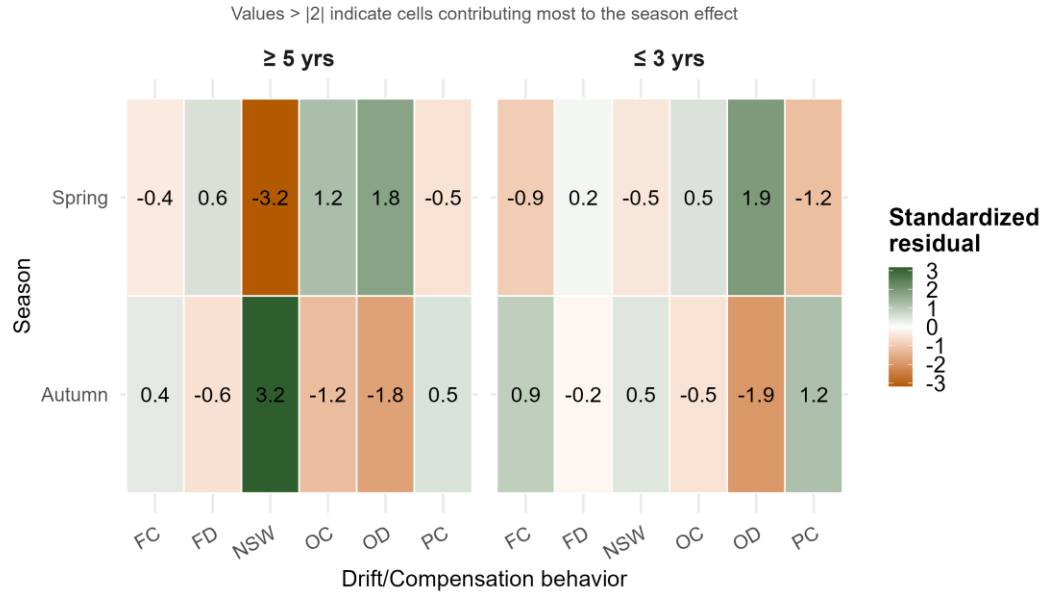
Relationship between the parameters of the cross-country optimization strategy.
Cross-country behavior was characterized by fitting horizontal inter-thermal gliding speed, vxy, as a function of thermal climb speed, vThermal climb, following vxy = AObs·vThermal climb + BObs. (a) Relationship between AObserved and BObserved across individuals (n = 11; $R^2 = 0.923$, $p < 0.001$). (b) Cross-country strategy lines vxy = AObs·vThermal climb + BObs for a range of AObs and BObs values following the relationship in (a), color-coded by AObs value; the dashed circle marks the common intersection point (x = 14.7, y = 1.81), determined from the intercept and slope of the fit in (a). Inter-thermal horizontal speed, vxy, is plotted on the x-axis and climb speed, vThermal climb, on the y-axis. (c) Cross-country strategy lines for each individual, using the same axes and color scale as in (b). (d) Cross-country strategy lines for all individuals combined (AObs = 0.718, BObs = 13.76, $R^2 = 0.125$, $p = 0.002$), and separately for experienced (AObs = 0.483, BObs = 14.79, $R^2 = 0.098$, $p = 0.120$) and inexperienced individuals (AObs = 0.837, BObs = 13.23, $R^2 = 0.142$, $p = 0.008$).

Appendix 9: *Wind drift/compensation behaviors, chi-squared test*

a



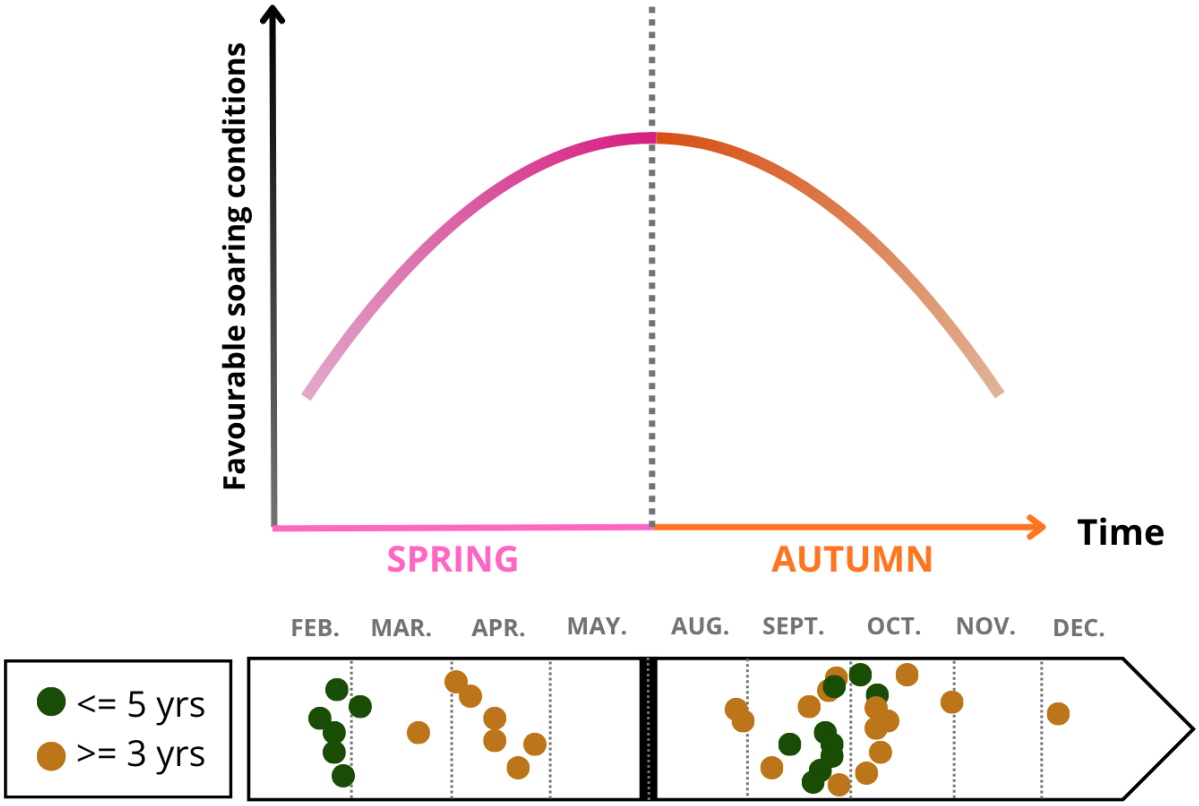
b



Chi-square tests of association between compensation category and experience / season. Standardized residuals from the contingency tables (a) remained below |2| for all behavioural categories in spring. In autumn, only the no-significant-sidewind category (NSW) approached this threshold (std. residual = 2.7 for experienced, −2.7 for inexperienced individuals), suggesting a marginal tendency for experienced individuals to be slightly overrepresented in segments with no significant sidewind. Standardized residuals (b) remained below |2| for all categories in inexperienced, consistent with the absence of a significant association. In experienced, the no-significant-sidewind category (NSW) was the main contributor to the season effect, with a standardized residual of 3.2 in autumn and −3.2 in spring, indicating that experienced individuals encountered NSW conditions disproportionately more often in autumn than in spring. No other category exceeded |2|.

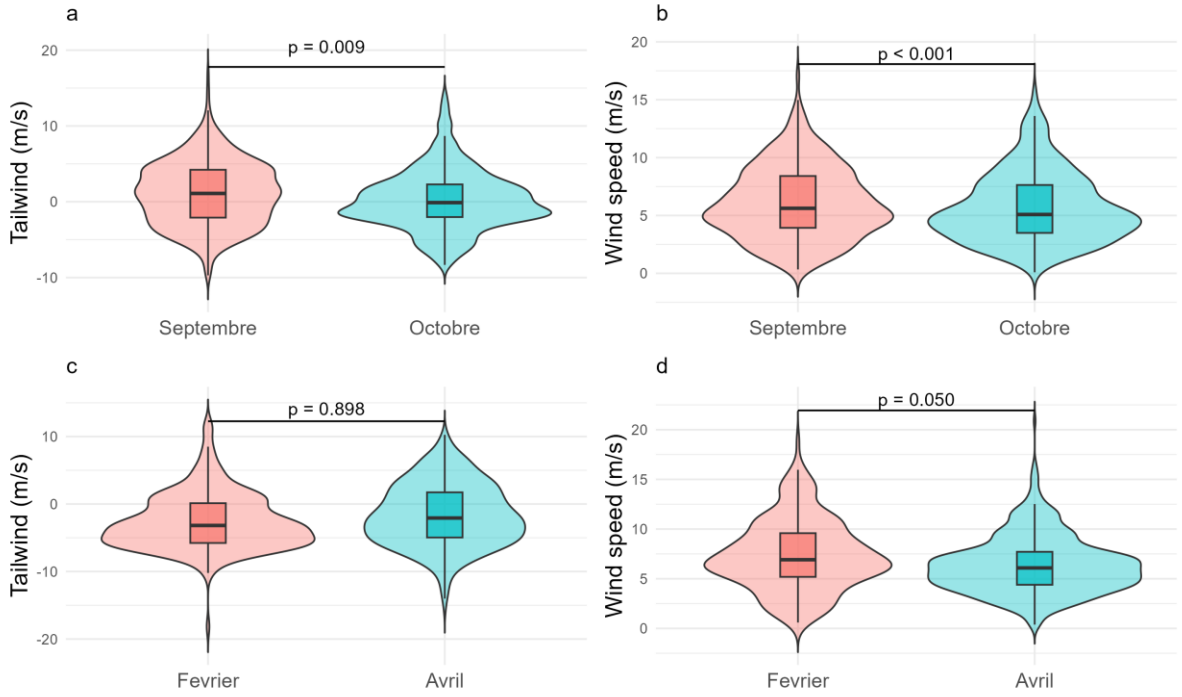
Appendix 10: *Migration periods and comparison of wind conditions*

Appendix 10.1: *Migration periods and soaring conditions*



Dates of entry into the study zone (Sahara) of experienced individuals (green) and inexperienced individuals (orange). The main migration month is February for experienced individuals in spring versus April for inexperienced individuals, and September versus October in autumn. **On the upper section, a schematic representation of the evolution of favourable conditions across the season** (improving as the season progresses in spring, and deteriorating as the season progresses in autumn). This illustrates that experienced individuals migrate under better conditions than inexperienced individuals in autumn, and conversely in spring.

Appendix 10.2 : *Comparison of wind conditions*



Comparison of wind conditions (tailwind and wind speed) between months within the autumn migration period (September vs October, panels a-b) and the spring migration period (February vs April, panels c-d). Violins show the distribution of hourly values, with boxplots overlaid (median, quartiles). Linear mixed models (wind variable ~ month + (1|bird/track)) show that in autumn, tailwind was significantly weaker in October than in September (estimate = -1.42 m/s, p = 0.009), as was wind speed (estimate = -1.64 m/s, p < 0.001). In spring, tailwind did not differ significantly between February and April (estimate = -0.27 m/s, p = 0.898), whereas wind speed was weaker in April (estimate = -1.11 m/s, p = 0.050). For each comparison, linear mixed-effects models were fitted using hourly observations of tailwind and wind speed. The analyses included 1,058 observations from 13 birds and 22 migration tracks for September–October, and 731 observations from 12 birds and 17 migration tracks for February–April.

Internship Overview

Beyond the scientific work presented in this thesis, the internship was an opportunity for numerous professional experiences, exchanges with a network of stakeholders in raptor conservation and research, and complementary fieldwork missions.

Meetings and scientific exchanges

The internship allowed direct exchanges with several researchers and experts in raptor telemetry tracking, notably Tonio Schaub, who provided invaluable support with GPS data preprocessing. Raymond Klaassen was particularly helpful with the sections concerning wind drift/compensation behaviors, allowing me to read his student's work on the topic and benefit from his expertise and sources. The question of identifying flight types (thermal soaring versus gliding) led to discussions on different possible approaches, notably with Benoît Van Hecke, who is developing an interactive website for tracking flight types from GPS position files (Flighttrace : https://birdtrack.netlify.app). We exchanged ideas both to explore this approach and to help improve the site along the way. Similar methodological exchanges on this topic also took place with Arzhela Hemery, who also studies flight behaviour as part of a power line avoidance programme and was working on developing an algorithm for flight identification. Direct communication was additionally established with Göksel Keskin, whose work informed the flight phase classification methodology used in this thesis. Beyond these individual collaborations, I sought to integrate more broadly into the scientific community by attending several scientific and outreach events during the internship: the IMBE general assembly, the Journées Montagne, the Cafés de l'Arbois, and the Journée des Doctorants (JDD).

Fieldwork and engagement with a network of stakeholders

I took part in observation sessions at sites where nests had previously been documented, to check whether they were being reoccupied at the start of the season, as well as in sessions dedicated to spotting and attempting to locate new, previously undocumented nests. I also devoted time to observing the flight behaviour of Short-toed Snake Eagles, as well as of other species such as Griffon Vultures (*Gyps fulvus*), in order to draw comparisons regarding the spacing between successive thermal uptakes, the time spent within thermals, and the duration of a single circling turn. Fitting GPS tags on adult Short-toed Snake Eagles at the Buoux wildlife care centre was also part of my involvement, along with tagging juveniles directly at the nest. Indeed, the internship was carried out in close collaboration with the LPO Buoux

wildlife care centre (especially Loriane Aubinais), the Lançon ZPS (Julie Corbon), and the ZPS of the Grand Site Concors Sainte-Victoire (Alexandre Lautier), which allowed engagement with a network of local raptor conservation stakeholders and access to their field expertise. It also involved a team of field technicians, notably Christian Marchal, a climbing field technician responsible for retrieving juveniles from the nest for tag deployment. Finally, I also took part in collecting Short-toed Snake Eagle feathers found beneath the nest, which will undergo genetic analysis to support population monitoring of the species.



Fitting a GPS tag on an adult Short-toed Snake Eagle, wearing a falconry hood, at the Buoux wildlife care centre.



Fitting a GPS tag on an adult Short-toed Snake Eagle at the Buoux wildlife care centre, by Alexandre Millon.



Juvenile tagged directly at the nest by Alexandre Lautier.

Additional missions

In addition to the main focus of the internship, I helped fit a tag on and release an adult Montagu's Harrier (*Circus pygargus*) at the Buoux wildlife care centre. I was also involved in a Little Owl (*Athene noctua*) conservation programme in the southern Luberon, which included handling chicks, recording their body measurements for ringing, and reading data loggers installed on the nest boxes. Finally, I assisted in observing a Golden Eagle (*Aquila chrysaetos*) nest for a morning, an observation session aimed at assessing the female's activity around the nest in order to help determine whether incubation was underway. This work also gave me the opportunity to meet other stakeholders involved in the conservation of different species, notably members of the Woodchat Shrike (*Lanius senator*) conservation programme (notably Olivier Hameau), broadening my understanding of the local raptor and farmland-bird conservation landscape beyond the scope of the main study species.